Technical Supplement

Engineered Intelligence: Tracing Computational Design from Search to Evidence

Research draft, 20 October 2026. Author approval pending.

This supplement retains the detailed representation, score, numerical-method, and cohort inventories supporting the main paper. Original evidence and contracts retain their own producer identities. It reports computational records; physical observations are not reported in this study.

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I. Historical Representation and Search Details

This supplement binds detailed decoding and variation statements to historical producer 82f450f7. The historical accession receipt binds qd.rs at that revision. The associated genome.rs and pheno.rs are inspected source, not evidence that every other cohort used their rules or that the original numerical work was replayed. Manuscript source and evidence-producer revisions remain separate identities.

A. Schema and Literal Producer Domains

The numeric groups occupy indices $0..39$, $40..89$, $90..116$, and $117..127$; the categorical material field follows them. The historical population CSV has 17 metadata/metric columns followed by these 129 genome-and-material fields. Its allele names and bounds are preserved below directly from the producer. A listed domain is a stored-coordinate domain, not an assertion that every value yields a distinct active geometry.

The silhouette bounds are symmetric about zero. Stations 0, 39, and 19–22 are pinned; stations 1–7, 14–18, and 23–27 allow ± 3.0 mm; station 8 and stations 28–29 allow ± 2.0 mm; stations 9–13 and 30–35 allow ± 3.5 mm; stations 36–38 allow ± 2.5 mm.

In the literal tables, T_MIN=5.3 mm, P1_POCKET_MAX=0.6 mm, and P1_RUNOUT_MIN=3.0 mm. The stock ceiling t_max() depends on JARAMBIT_T_MAX, constrained between the floor and 6.0 mm, defaulting to 6.0 mm. The source table alone does not establish a particular run's environment value. Units and threshold/floor interpretation follow the named decoder; source identifiers are retained to avoid inventing unit assignments for ratios and selectors.

TABLE I
Historical literal coordinate domains, indices $40..55$.

Index	Source name	Stored bounds
40	blade_drop_deg	[-4.0,4.0]
41	grip_dx	[-0.4,10.0]
42	ring_wall	[4.5,7.0]
43	ring_id	[24.9,25.9]
44	neck_fillet	[8.0,18.0]
45	t_stock	[T_MIN,t_max()]
46	t_tip	[2.6,4.5]
47	taper_x0	[115.0,150.0]
48	edge_angle_in	[14.0,22.0]
49	edge_angle_sp	[14.0,22.0]
50	comet_x0	[22.0,30.0]
51	comet_x1	[56.0,70.0]
52	comet_w0	[7.0,12.0]
53	comet_w1	[3.5,6.0]
54	comet_bow	[0.0,4.0]
55	comet_y	[0.5,4.0]

B. Clamping and Decoded Types

The historical clamp replaces nonfinite values with lower bounds, applies box bounds, and quantizes stock through the bound q01 routine. It enforces $t_{\mathrm{tip}} \le t_{\mathrm{stock}} - 0.5$ mm, holder-wall and engagement constraints, a fuller interval separation of 0.15 (capped at 0.9), a comet span of at least 28 mm, and a window interval separation of 0.2 (capped at 0.92). These are producer-specific rules, replacing the manuscript's earlier unbound generic examples.

The continuous-rail layer activates at p1_on ≥ 0.5, or through its environment override. It disables selected legacy pocket/through-feature controls, caps shallow pockets, and bounds thickness knots. With the layer off, all stored knot values are overwritten by stock thickness during clamping. Active knot count is floor(k_n) bounded to 3–5. The phenotype uses floor(win_n) for window count and a threshold at 0.5 for the texture mode. It does not round every selector to an integer in the serialized vector. Thresholded zone controls and inactive values further separate the number of stored coordinates from effective geometric changes.

TABLE II
HISTORICAL LITERAL COORDINATE DOMAINS, INDICES $56..71$.

Index	Source name	Stored bounds
56	eye_scale	[0.0,1.25]
57	cairn_d	[4.0,7.0]
58	nub_engage	[0.4,1.2]
59	holder_wall	[1.2,2.0]
60	jimp_scale	[0.8,1.2]
61	serr_large_d	[1.0,1.5]
62	serr_small_d	[0.4,0.8]
63	serr_fine_d	[0.5,0.9]
64	spine_sharp_offset	[0.0,45.0]
65	relief_d	[0.0,1.2]
66	relief_w	[2.0,8.0]
67	relief_q	[1.0,3.0]
68	lattice_d	[0.0,1.5]
69	lattice_pitch	[4.0,9.0]
70	lattice_frac	[0.4,0.8]
71	lattice_angle	[0.0,60.0]

TABLE III
HISTORICAL LITERAL COORDINATE DOMAINS, INDICES $72..87$.

Index	Source name	Stored bounds
72	lattice_handle	[0.0,1.0]
73	lattice_blade	[0.0,1.0]
74	fuller_d	[0.0,1.2]
75	fuller_w	[2.0,5.0]
76	fuller_s0	[0.05,0.4]
77	fuller_s1	[0.5,0.9]
78	flute_d	[0.0,1.0]
79	texture_mode	[0.0,1.0]
80	texture_d	[0.15,0.4]
81	texture_pitch	[1.5,3.0]
82	texture_angle	[-45.0,45.0]
83	wrap_x0	[16.0,26.0]
84	wrap_len	[45.0,60.0]
85	pad_x	[70.0,80.0]
86	pad_y	[-5.0,3.0]
87	pad_w	[6.0,9.0]

C. Crossover and All Six Mutation Branches

Historical crossover chooses the 40-coordinate silhouette block from either parent with equal probability, then chooses every other numeric coordinate and the material category independently with equal probability, followed by clamping.

For mutation, op=floor(6u) selects a mode from 0–5 using the producer RNG. The supplied scale is σ; fine means $\sigma < 0.025$, and silhouette scale becomes 0.1σ in that branch. Let s_i denote coordinate-domain span. A jig adds a normal variate multiplied by the supplied scale and s_i.

1) Mode 0: attempt each coordinate with probability 0.15, using silhouette scale for the first 40 and σ otherwise; stock is omitted in fine mode.
2) Mode 1: choose a periodic silhouette-bump center in $[0, 40)$, width from $[1.5, 5.0]$, and normal amplitude scaled by 6 times the silhouette scale. Apply the source's exponential profile with station-bound scaling. Outside fine mode, also attempt drop and grip-offset perturbations, each with probability 0.3.
3) Mode 2: attempt thickness, taper, relief, bevel, ring-wall, and edge-thickness controls with probability 0.5 and jig scale 1.5σ, excluding stock in fine mode. When the rail layer is active, attempt each of five knots and knot count with probability 0.5.

TABLE IV
HISTORICAL LITERAL COORDINATE DOMAINS, INDICES 88..103.

Index	Source name	Stored bounds
88	pad_l	[12.0,16.0]
89	edge_t	[0.5,0.9]
90	iso_d	[0.0,1.6]
91	iso_pitch	[7.0,16.0]
92	iso_rib	[1.4,3.0]
93	iso_angle	[-30.0,30.0]
94	iso_rim	[2.0,5.0]
95	iso_blade	[0.0,1.0]
96	iso_handle	[0.0,1.0]
97	iso_blade_extent	[0.0,1.0]
98	iso_grade	[-0.8,0.8]
99	win_n	[0.0,7.99]
100	win_s0	[0.05,0.5]
101	win_s1	[0.5,0.92]
102	win_w	[3.0,10.0]
103	win_fill	[0.45,0.8]

TABLE V
HISTORICAL LITERAL COORDINATE DOMAINS, INDICES 104..119.

Index	Source name	Stored bounds
104	win_decay	[0.4,1.0]
105	rflute_d	[0.0,2.0]
106	rflute_w	[0.3,0.7]
107	rflute_arc	[120.0,260.0]
108	iso_handle_start	[0.0,0.8]
109	iso_land	[0.0,1.0]
110	perf_d	[0.0,7.0]
111	perf_gap	[3.2,8.0]
112	perf_angle	[-40.0,40.0]
113	perf_elong	[1.0,2.4]
114	perf_land	[0.0,1.0]
115	perf_handle	[0.0,1.0]
116	perf_handle_start	[0.0,0.9]
117	p1_on	[0.0,1.0]
118	rail_w	[3.0,7.0]
119	neck_l	[4.0,12.0]

TABLE VI
HISTORICAL LITERAL COORDINATE DOMAINS, INDICES 120..127.

Index	Source name	Stored bounds
120	pocket_d	[0.0,P1_POCKET_MAX]
121	pocket_r	[P1_RUNOUT_MIN,6.0]
122	k_n	[3.0,5.99]
123	k0	[T_MIN,6.0]
124	k1	[T_MIN,6.0]
125	k2	[T_MIN,6.0]
126	k3	[T_MIN,6.0]
127	k4	[T_MIN,6.0]

4) Mode 3: attempt lattice depth, pitch, fraction, and angle with probability 0.6 at scale 1.5σ. Outside fine mode, flip each handle/blade activation coordinate with probability 0.2.
5) Mode 4 (the remaining match branch): attempt fuller, flute, texture, comet, eye, and cairn controls with probability 0.4 at scale 1.5σ; omit texture mode in fine mode.
6) Mode 5: with the rail layer active, attempt rail, neck, pocket, knot, stock, fuller-depth, ring-wall, and neck-fillet controls with probability 0.4 at scale 1.5σ. Otherwise attempt the source's machining-coordinate list with probability 0.4; outside fine mode, flip five zone controls with probability 0.12 and attempt a ± 1 window-count step with probability 0.25. Stock is omitted in fine mode.

Outside fine mode the material category flips with probability 0.03. Every mutation ends in clamping. The historical island loop supplies island/configuration-dependent scales, including a 0.6 multiplier after crossover; the earlier manuscript's fixed 0.15/0.04 pair did not describe this bound loop. Source lists remain available in the immutable genome implementation.

D. Membership Stage and Frame Interpretation

The bound island loop constructs novelty context, refreshes fitness, emits birth/elite membership events and summary observations, and calls its snapshot callback before migration. Migration copies selected IDs to the next island. A GenEnd event is emitted after that migration, but its summary values were gathered before it. Event position and the stage summarized by its values are therefore different facts. JEV1 elite events omit refreshed scalar fitness. The reporting adapter accepts complete and maps historical notevaluated to not_evaluated; it rejects unknown status strings.

E. Inheritance Ledger and Interpretation Boundary

Two source-bound reporting paths coexist. The original lineage-inputs.v1.json defines a two-hop view consumed by lineage::project, which produces generated/lineage.json and generated/lineage.svg. Separately, the population reporter reconstructs generated/population/inheritance.json directly from the hash-pinned historical all_evals.json and all_evals.csv. Its validator checks the seven three-hop parent edges and separately identifies the recorded fourth-hop induced closure $174 \rightarrow 299$; this path produces figures/inheritance.svg, the ancestry view used here. The ledger also retains exact numeric-coordinate differences, material change indicators, and mixed-unit raw norms. The two-hop contract remains unchanged and is not the authority for the separate three-hop view. Figure 1 retains the detailed coordinate-change annotations omitted from the compact main-paper view.

II. Scores, Candidate Roles and Display Composites

A. Original Reporting Authority and Missing Provenance

The six sources admitted by docs/paper/weighted-inputs.v1.json contain fifteen generation summaries. That original contract leaves producer revision, scorer revision and point stage unestablished. The original score-profile registry names a producer tuple for the weighted profile: revision 009f5ac520fa3afe8ed8025230eef604dbb7f922, path crates/paper-artifacts/src/validation.rs. That path is absent at the named revision. A compatible versioned recipe is nevertheless retained at ops/ci/ultralight_weighted_fitness.py at that revision, SHA-256 1adbd37191710d439d3ca3 88e468499fe48bc217546145deb57f87f1da24770b. Its compute_ultralight_weighted_fitness function defines the twenty weights and normalizations. This recovers a recipe-source binding, not proof of the exact code or observation stage executed for each historical summary. The original registry tuple and evidence bytes remain preserved; absence of that code at another recorded campaign revision alone does not establish when the summary was computed.

The source for Individual 2002's vector is evidence/evolution_waves/2026-10-18_ultralight_v1/xbabe3/w411c3/fit ness_curve.json, SHA-256 afe065170218f0f5f20942a9e737e7e91a300244f8e6a99fd3c1d617d16f1412, selector /best_so_far/ultralight_fitness_breakdown/objectives. The scalar is 74.207299. The weights and normalized terms below are serialized summary values, not recomputed solver observations. The exact executed historical recipe and point-stage binding remain unestablished. The spatial display's within-channel ranks, available-channel mean and color transfer are separate from the recovered weighted-score recipe. The saved terms' weighted-point sum reproduces the saved scalar to retained precision. A separate certificate-matched leaderboard retains Individual 2002's later-assessment summary (Supplement Section V); it does not supply its missing underlying CAD/full scorecard artifacts. No such artifacts are inferred from the registry category, certificate hash or component labels.

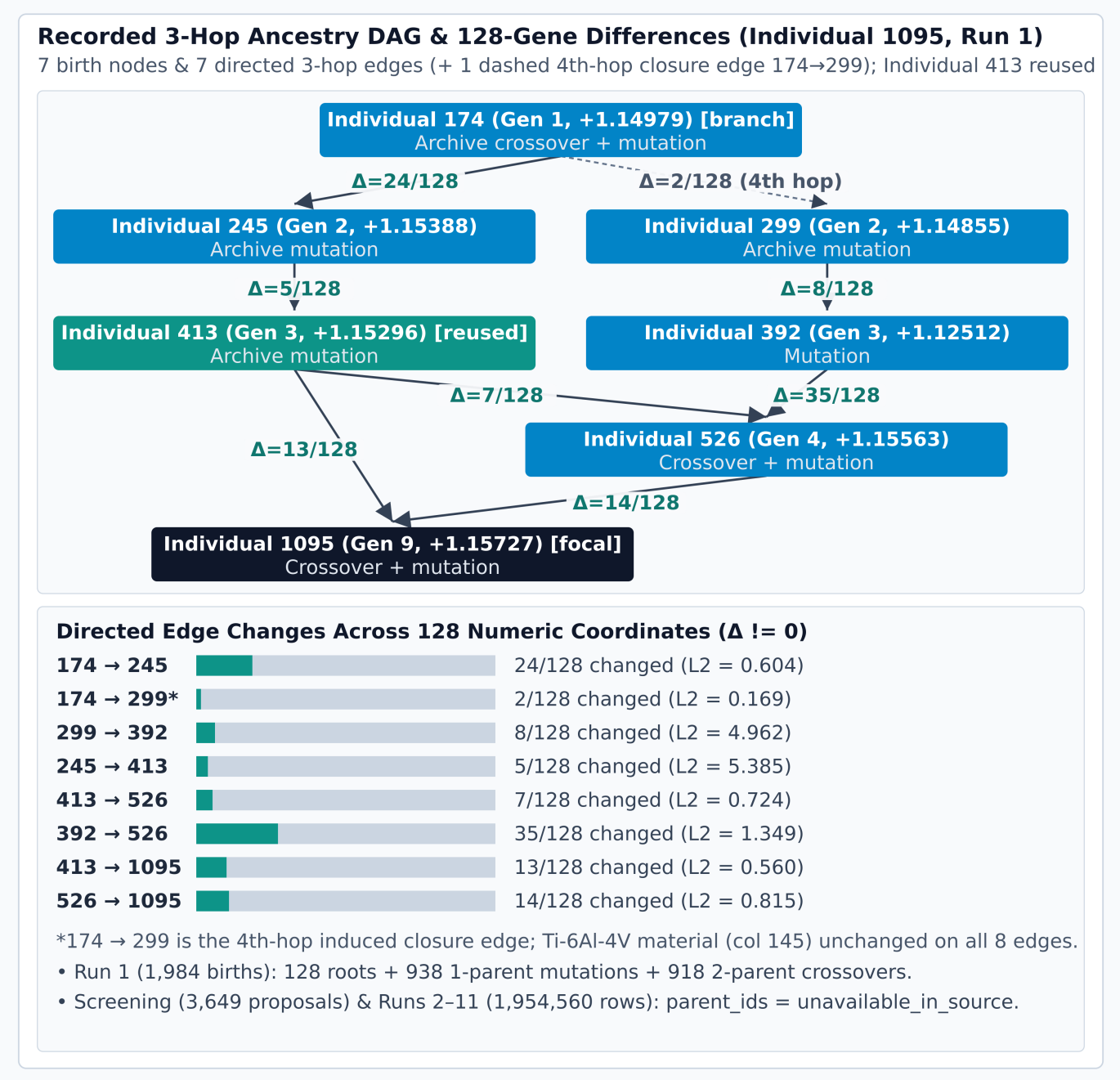


Fig. 1. Detailed retained ancestry annotations around Individual 1095. The raw coordinate norms mix parameter units and are descriptive stored-coordinate differences, not a physical or normalized genetic distance. Source-coordinate differences, parent ordinals, material indicators and the induced fourth-hop edge remain in the unchanged inheritance ledger.

B. Default-Compatible Terms and Execution Limits

The saved legacy input breakdown contains bore-edge radius $1.125\,\mathrm{mm}$; edge apex/half-angle $0.25\,\mathrm{mm}/28°$; point thicknesses $0.45/0.85/1.40\,\mathrm{mm}$ with a false monotonicity flag; and overload factors $0.10/0.35$ for torsion/spine. These exactly match the recovered recipe's fallback constants when the corresponding measurements or favorable flags are absent. They are default-compatible reported inputs, not established measurements. The original call inputs and exact executed-source binding are incomplete, so equality to a constant alone does not prove the branch taken for every historical value.

The four whole terms bore/collar, edge, point and overload contribute 7.875000, 4.232143, 2.608543 and 0.966667 points, totaling 15.682353. That total includes the whole bore/collar term, whose bearing-width half is separate

TABLE VII
INDIVIDUAL 2002'S ACTUAL RETAINED WEIGHTED VECTOR: SCIENTIFIC LABELS PLUS EXACT MACHINE KEYS. NORMALIZED VALUES AND CONTRIBUTIONS ARE SAVED REPORTING QUANTITIES.

Objective / machine key	**Weight (%)**	**Normalized**	**Points**
Mass preference; ultralight_mass	22.0	0.339548	7.470061
Holder/compactness proxy; holder_and_compactness	8.0	1.000000	8.000000
Ring/claw form proxy; karambit_ring_and_claw	7.0	1.000000	7.000000
Reference form proxy; c6_aesthetic_and_ergo	9.0	1.000000	9.000000
Bore/collar comfort proxy; bore_collar_comfort	9.0	0.875000	7.875000
Single-solid B-rep status; cad_watertight_brep	5.0	1.000000	5.000000
Fallback-compatible edge-profile proxy; gate_s_sharpness	5.0	0.846429	4.232143
Fallback-compatible point-profile proxy; gate_s_point_profile	5.0	0.521709	2.608543
Certification lateral, 180 N; hiker_lateral_180n	3.5	1.000000	3.500000
Certification lateral, 260 N; hunter_lateral_260n	2.5	0.708077	1.770192
Strict lateral, 260 N; strict_hunter_260n	2.0	0.757692	1.515385
Ring traction, 1500 N; ring_pull_1500n	2.5	1.000000	2.500000
Body torsion, 1.82 N m; twist_body_1p82nm	2.5	1.000000	2.500000
Local plate torsion, 1.82 N m; twist_tip_plate_1p82nm	2.5	0.260000	0.650000
Reported solid-tip/comparison proxy; solid_tip_3d_p2_and_xcheck	3.0	0.363600	1.090800
Reported support/stability proxy; side_a4_buckling	2.0	1.000000	2.000000
Elastic yield-order proxy; failure_sequence_1p2	3.5	1.000000	3.500000
Reported plate-study status; plate_gci_3level	2.0	1.000000	2.000000
Overload proxy; human_max_overload	2.0	0.483333	0.966667
Goodman fatigue proxy; goodman_fatigue_l7	2.0	0.514254	1.028508
Total	100.0	—	74.207299

from its fallback-compatible edge-radius half; it is not 15.682353 points of proven pure imputation. Favorable missing-key defaults also occur for selected Boolean objectives. Those historical defaults qualify claims about missing data: the present reconstruction adds no invented observations, while preserving the saved score and its original proxy/default-compatible terms.

The versioned recipe also contains a latent undefined-name error in the legacy sharpness predicate: half_angle _max_deg is used where the local variable is half_angle_deg. The saved apex $0.25 > 0.20\,\mathrm{mm}$ short-circuits that Python and branch, so the defect alone does not preclude generating this candidate's vector. Current source corrects the name. Neither compatibility nor that correction proves which source bytes executed historically; no scorer replay or replacement score is introduced here.

C. Conditional CAD Score and Global Composite

At producer revision e8cb049df7d03c7d3113556812cc140ca99a29c3, ops/ci/evaluate_cad_candidate.py defines the CAD score's two branches:

$$W_{\mathrm{CAD}}^{\mathrm{pass}} = 1 + 0.70 s_{\mathrm{mass}} + 0.12 s_{\mathrm{form}} + 0.10 s_{\mathrm{survival}} + 0.05 s_{\mathrm{seq}} + 0.03 s_{\mathrm{carry}}, \tag{1}$$

$$W_{\mathrm{CAD}}^{\mathrm{fail}} = -\left(1 + 4 \sum_{k} \Delta_k\right) + 0.10 s_{\mathrm{mass}}. \tag{2}$$

The pass branch applies only if all sixteen legacy predicates are true. The mass subscore is a clipped mixture of linear and logarithmic rewards between the 92 and 74 g references. Form combines reference score, hook-drop and curvature proxies. Survival and sequence terms use saturating margin functions; the carry term combines the saved support eigenvalue and fatigue factor. These historical names and weights describe the score's preference rule; they do not establish the physical interpretation implied by a component name.

For Individual 2000, the bound scorecard saves $s_{\mathrm{mass}} = 0.486999$, $s_{\mathrm{form}} = 0.400862$, $s_{\mathrm{survival}} = 0$, $s_{\mathrm{seq}} = 0.999604$ and $s_{\mathrm{carry}} = 0.6$. Its nonzero normalized shortfalls are strict lateral 0.3530769231, strict tip torsion 0.471 and solid tip 0.3307. The saved penalty is 4.619108 and the actual failed-branch score is -5.570408. The saved unpenalized value 1.456983 is hypothetical for this failed candidate. Favorable legacy flags such as plate_gci_valid_3level remain historical bookkeeping; they do not override the refused extrapolated-error estimate. The original support eigenvalue and pass flag are likewise preserved, but are not admitted as validated buckling performance. Their contribution to a saved score is not recomputed or promoted to new mechanical evidence.

The separate dimensionless composite is

$$\begin{aligned} G_1 = {} & 0.35 \min(\mathrm{SF}_{\text{solid tip}}, 2) + 0.35\,\mathrm{SPG}/4 + 0.20 s_{\mathrm{shape}} \\ & - 0.05 \max(0, m_g - 77)/10 \\ & - 0.05 \max(0, 1.2 - r_{\mathrm{seq}})/1.2. \end{aligned} \tag{3}$$

The original metrics/global_metric.py requires valid inputs rather than filling missing terms with zero. A screen-tip substitute is explicitly provisional. A complete G_1 status concerns input completeness; it does not mean the candidate satisfies all requirements. Individuals 2001/2007 retain $G_1 = 0.623903/0.621634$ and $W_{\mathrm{CAD}} = -3.878704/-3.964924$ under their separate controlled-export records.

D. Native Context and Display Transform

The historical native scorer is bound to 82f450f7fb418e28c4b229031920dec5454435f3, crates/crescent/src/qd.rs, SHA-256 fe3d057eeec74c3278b46027265406ec0ee97dd65938ae49cb003e08ee5b3a5f. Illegal/infeasible branches can be negative; evaluated records add mass/mechanical terms and contextual novelty. Novelty uses contemporaneous descriptor context and is re-evaluated during search. In historical Run 1, the 446 legal births have positive stored fitness and the other 1538 do not. In the distinct completed native cohort, 610,051 search-legal births have negative contextual fitness. The sign/legality relation therefore requires its cohort and producer context. Stored birth observations, rescored memberships, generation/island summaries and cumulative archive representatives remain distinct stages.

Negation $L = -f$ or $-W_{\mathrm{CAD}}$ and complement $L = 100 - F$ reverse numerical order while preserving preference under minimization. A cumulative maximum $H_g = \max_{k \le g} r_k$ is nondecreasing by construction; its loss transform is nonincreasing. These transforms preserve a profile's preference, not comparability between profiles or evidence of physical improvement.

E. Complete Roster and Missing-Aware Display Recipe

The additive docs/paper/data/test-field-roster.v1.json inventories all 44 catalog predicates (16 G16, 10 RSV2 and 18 battery rows) plus 8 original-channel metadata rows. The resulting 52 rows are an inventory, not independent experiments or a count of performed solves. The historical 23-subject battery remains a separately bound accession. Every row retains its source identity, quantity, units, observation stage, domain, dependence and missingness reason. Related D1 response maps do not supply exact spatial observations for certification, strict/refined or regionally recovered scalar predicates.

The contract docs/paper/data/display-composite-contract.v1.json admits four canonical channels on Individual 2000's original $25{,}209$ D1 triangles: /mesh/lat and /mesh/pry in MPa/N, /mesh/ring in MPa/N, and raw /mesh/twist in MPa/(N mm). Pry is an additive source/figure-semantics admission; original input contracts remain unchanged. The scaled maximum envelope and its derived safety-factor array are dependent companions, excluded from another equal-weight contribution. Geometry/thickness arrays are excluded too. Missing refined, local-solid, eigenvector and fatigue maps, scalar-only predicates and foreign-domain observations contribute null with a reason; no coarse-map substitution or scalar broadcast supplies them. Individual 2002's distinct genome, certificate and weighted reporting vector supply no fields for Individual 2000.

Eligibility requires the original triangle association, finite values and positive original triangle area. Ring's producer half-plane mask, $x_{\mathrm{centroid}} \ge s_{x0} - 3\,\mathrm{mm}$, where s_{x0} is the producer's seat-start coordinate, includes the seat and unloaded blade; it is derived from the retained mesh/genome and checked against the admitted mask fingerprint. Its $14{,}875$ masked cells are excluded. A zero outside that explicit mask is valid. Lateral, pry and raw torsion have no added value-based masks; the capacity exclusion is not a mask on raw torsion. Missing values remain typed nulls

with reasons. Nonfinite/untyped values, unexpected absent keys, identity/hash or registration mismatches are errors, rather than silently skipped values. Zero-area cells remain inventoried but contribute neither rank nor mean; an empty or zero-total-area channel has no normalized values.

For channel k, write $A_{k,<v}$ for valid source area at values strictly below v, $A_{k,=v}$ for the pooled area exactly tied at v, and $A_{k,\mathrm{valid}}$ for its total valid area. The ascending response rank is

$$r_k(v) = \frac{A_{k,<v} + A_{k,=v}/2}{A_{k,\mathrm{valid}}}. \tag{4}$$

Exactly equal serialized values, including signed zeros, receive the same rank regardless of source-cell order. A constant positive-area field therefore has rank 0.5. Each channel retains its raw units/range, source hash, mask, valid area, counts and tie policy; these ranks do not reconstruct the historical weighted-fitness normalization.

For the set K_i of available canonical normalized channels at cell i, define $q_i = |K_i|$ and $c_i = q_i^{-1} \sum_{k \in K_i} r_{ki}$ when $q_i > 0$. If $q_i = 0$, the index is null and the contributor list is empty. This is an unweighted mean of available finite values, with no post-mean candidate min/max stretch. Per-cell records retain contributor IDs, count and missing reasons. The support denominator is four eligible channels, never 44 predicates or 52 metadata rows. Coverage uses source triangle area before projection or decorative locators: channel support and the area fractions for $q = 0, \ldots, 4$ use the whole positive source-mesh area as denominator. A high index with one contributor must remain distinguishable from the same index with four; none denotes safety, scientific fitness or desirability.

The defined full-range view maps $c_i \in [0, 1]$ through a versioned navy–teal–yellow–orange–red transfer with shared surface/key code, direct piecewise sRGB interpolation and no simulated lighting. Its luminance is not claimed monotone. Numeric ticks and explicit support supply additional cues. The contract also defines an area-weighted 5th–95th percentile contrast preview, not a selected replacement for the full-range view. If that preview is used, its key must report bounds in the original index, with counts and source-area fractions strictly below/above them and visible saturation cues. The contrast coordinate is distinct from c_i; an empty/collapsed window must not create invented variation. The full-range companion remains required.

Surface colors project the plate index to the retained same-export STL by nearest original D1 triangle centroid in unposed XY, with exact ties resolved by source index. The original surface triangles, silhouette and holes remain unchanged. This display assignment is not a solved volumetric stress field. The predecessor five-map, source-order-tie, candidate-minmax recipe and its sensitivity receipts remain preserved as superseded display evidence; their contributor counts and differences are not results of this four-channel recipe. Actual generated values, support receipts, contrast choice and reading-size PNG/PDF acceptance require their matching output evidence.

F. Crop Measure, Denominator and Range

For a piecewise-constant stored triangle field v, the display-only saliency measure is $\int_D |v|\, dA$. A crop's numerator integrates over the actual triangle/bbox intersections, and its denominator integrates over all source-active triangles in the declared domain. A chosen prior ROI is a different denominator from the whole retained field. Coordinates use the proper rigid pose $R_z(-6^\circ)R_x(180^\circ)$.

The current source-bound SVG uses the posed rectangular domains $[-28, -26, 84, 23]\,\mathrm{mm}$ for ring traction, $[166, 4, 198.5, 37]\,\mathrm{mm}$ for terminal torsion and $[-20, -26, 196, 36]\,\mathrm{mm}$ for tip pry. Its explicit clip rectangles now match those physical bounds, rather than the earlier aspect-padded viewport. The generated docs/paper/generated/crop-semantics.v1.json binds the retained D1 source SHA-256 c2811ed8877904254ddc953dc888aedc3d7791b69c040dd6513fe0b763997bab to SVG SHA-256 4a048bb34fdccc2687f2fba63372af4a35868602522b1d3bf54782c44ae9ecb4. This is reporting geometry and bookkeeping, not a new solve or a change to field values or producer masks.

The whole source-active integrals are 14.173756, 119.158584 and 900.438870, over $10{,}334$, $25{,}209$ and $25{,}209$ active source triangles, respectively. Triangle-intersection crop integrals are 14.173756, 76.878008 and 900.438870, giving rectangular source-domain fractions 100%, 64.5174% and 100%. For B, 3033 emitted source triangles include 2474 that intersect the exact rectangle; neither count is the old 2062-cell ROI. These fractions are defined before decorative locator overlays and are not unobscured pixel/visible-ink fractions. They make no global 90% claim for B. The earlier literal-bbox calculation was diagnostic until polygon/affine/clip binding and the corresponding output were established; that pending status is superseded for this exact source-bound SVG. Independent PNG/PDF readability and final manuscript acceptance remain separate.

The superseded terminal receipt instead reports 2062 selected ROI cells (not an established count of currently emitted polygons) and integral 73.5323, compared with whole torsion integral 119.1586 over 25,209 cells. Its 100% ROI fraction is only about 61.71% of that whole-field denominator. Its exact prior ROI selection rule is unstated; it is not re-created or certified here. Preserve that receipt with explicit supersession rather than updating its hashes around a changed panel.

Raw torsion peaks at $34.8589\,\mathrm{MPa}/(\mathrm{N\,mm})$, inside the capacity exclusion; the post-exclusion capacity peak is $0.6611\,\mathrm{MPa}/(\mathrm{N\,mm})$. The bound B view uses the latter color-range endpoint, labels ≥ 0.6611 as over-range and prints the raw maximum 34.8589 separately; 59 source triangles intersecting its rectangle exceed the displayed endpoint. The color endpoint is not the raw maximum or a new capacity measurement. Panel C remains /mesh/pry, peak $2.9181\,\mathrm{MPa}/\mathrm{N}$, rather than the envelope in MPa.

III. Display Support and Additional Score Views

A. Complete Inventory and Spatial Availability

The complete row-level table is retained as docs/paper/generated/test-field-roster.v1.tsv, with its source-bound JSON contract in docs/paper/data/test-field-roster.v1.json. Every row gives its original key, scientific quantity, observation stage, availability and exclusion or missing-field reason. The JSON additionally binds source observations, hashes, domains, masks and dependencies. Table VIII reconciles its universe. Inventory membership does not imply an independent test, performed solve or spatial response. Scalar-only and incompatible-domain rows remain in the table and contribute missing values at every display cell; they are never broadcast. The requested multihue palette has no monotone-lightness or universal color-vision accessibility claim. Grayscale and simulated color-vision diagnostics can merge different index colors. Numeric per-cell values, contributor records and the separate physical-unit views remain necessary for quantitative interpretation; hue alone is insufficient.

TABLE VIII
Inventory reconciliation for Individual 2000's display. The four eligible response channels are metadata addenda, not four extra catalog tests.

Family	Rows	Spatial interpretation
Legacy G16	16	Retained predicates/scalars; related single-mesh fields do not duplicate threshold predicates.
Grouped RSV2	10	Grouped or scalar findings; no additional registered response arrays.
Embedded battery	18	Software, geometry, scalar or local-result records; no additional plate-response arrays.
Channel metadata	8	Four eligible response channels; two dependent derived companions; two geometry quantities.
Total metadata	52	44 catalog rows plus 8 channel rows, not 52 independent tests.

B. Quantitative Companions to the Appearance View

C. Source-Coordinate Crops and Whole-Field Denominators

Main Figure 6 changes only the viewing window. For each eligible original field, a range cutoff is $t_\alpha = v_{\min} + \alpha(v_{\max} - v_{\min})$ with fixed candidates $\alpha = 0.50, 0.75, 0.90$; it is not a count or area quantile. The baseline t_{50} always comes from the whole original eligible field, including when a more concentrated cutoff proposes the window. For original triangle areas A_i and their exact rectangularly clipped areas A_i^{crop}, the area and excess fractions are

$$\frac{\sum_i A_i^{\mathrm{crop}}}{\sum_i A_i}, \qquad \frac{\sum_i A_i^{\mathrm{crop}}(v_i - t_{50})_+}{\sum_i A_i (v_i - t_{50})_+}. \tag{5}$$

Both sums use the whole source-eligible set after its original masks; missing values are not zeros, and genuine finite zeros remain valid. Positive excess is display saliency, not physical energy. The saliency-selection policy requires at least 95% of that whole-field excess, a bounding-box fraction at most 0.40 and scale gain at least 1.75, with padding fraction 0.08. These are display constants, not physical or qualification gates; 95% is a policy threshold, not a reported achieved coverage. All threshold ties, maxima and disconnected support participate. Missing, constant or insufficiently concentrated fields retain full context; undefined fractions remain missing. No crop-local color

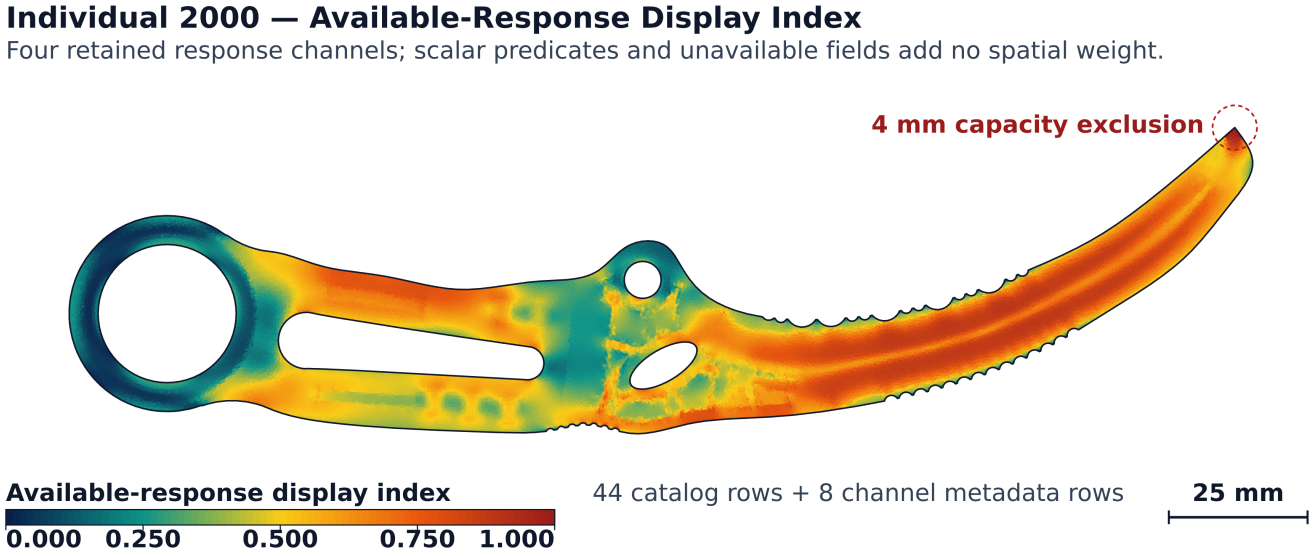


Fig. 2. Unlit full-range quantitative companion to the appearance view in main Figure 3. Surface and legend use the same flat transfer of the retained four-channel index over $[0, 1]$, with mesh off. Lighting changes RGB in the main appearance view; its colors must not be read as this quantitative key. Neither view changes the admitted geometry, responses, masks, contributor counts or retained scores, and neither establishes safety, fitness or qualification.

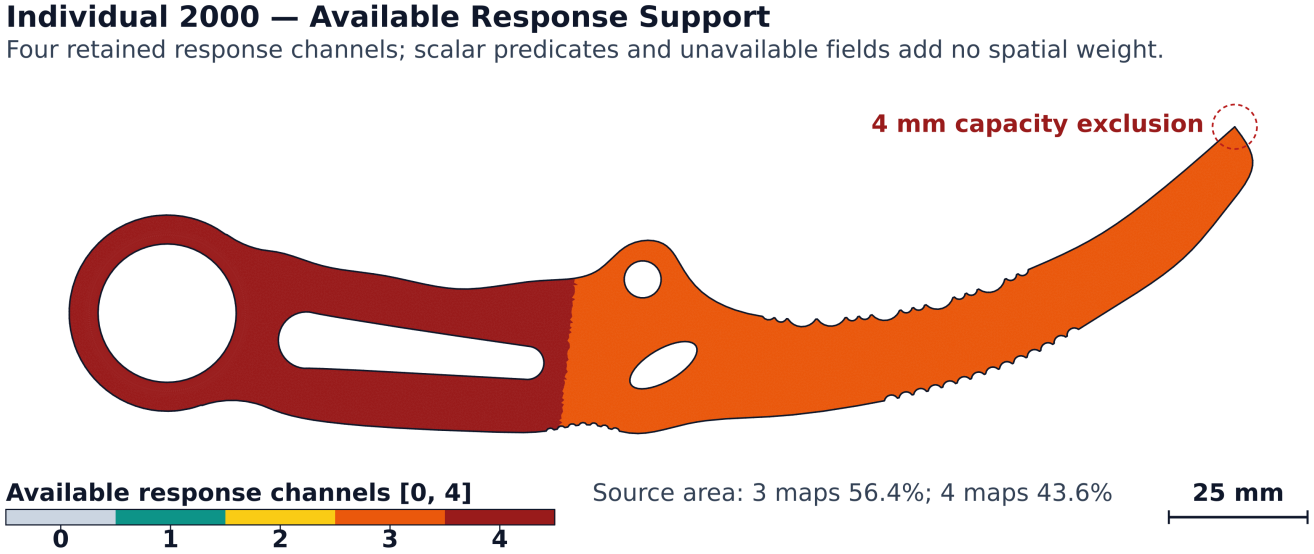


Fig. 3. Available-channel support for the selected full-range hero, before surface projection and decorative overlays. The denominator is the complete positive source-triangle area, $3816.272554\,\mathrm{mm}^2$. Support is $q = 3$ on 14,875 cells (56.4190% of source area), where the original ring mask excludes that channel, and $q = 4$ on 10,334 cells (43.5810%). No source cells have $q = 0, 1, 2$. Coverage is $q/4$, not $q/44$ or $q/52$; these are source-area fractions, not visible surface-ink fractions.

normalization is applied. For the lateral and torsional B/D panels, a separate display camera expands the selected ROI to at least 30% of the whole object's horizontal extent, with physical-aspect padding and clamping to the original source bounds. This wider camera shows curved distal blade/tip context. Its area and excess fractions are recomputed from the exact clipped original triangles; the smaller saliency-selection ROI retains its own threshold and statistics. The PiP red rectangle is the actual displayed camera, including this context, not the smaller selected ROI. The 0.40 compactness requirement above applies to the saliency selection, not to the expanded context camera. The source-bound docs/paper/generated/plate-smart-crops.v1.json records the actual decision, thresholds, source hashes, clipped areas, whole-field integrals and affine crop/locator coordinates for each panel.

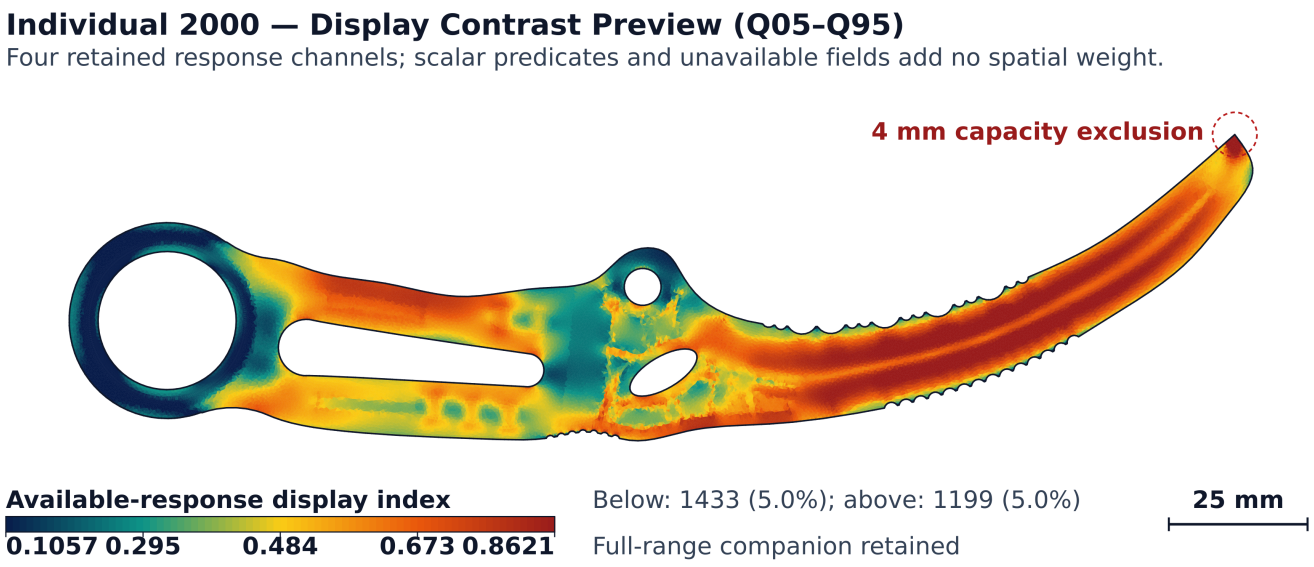


Fig. 4. Additional contrast preview, rather than the selected main rendering. The predeclared source-area-weighted 5th–95th percentile index window is $[0.1057475260, 0.8621273953]$. Below it are 1433 source cells over $190.720950\,\mathrm{mm}^2$; above it are 1199 cells over $190.751611\,\mathrm{mm}^2$. Each tail is approximately 5.0% of original *source area*, not 5.0% of cell count. Only this preview saturates those tails. Figure 2 retains the unlit full $[0, 1]$ quantitative key with mesh off; the main appearance view separately uses lighting. Colors are display coordinates, not physical units or fitness.



Fig. 5. Spatial concentration differs across Individual 2000's retained plate responses. A: scaled multi-case envelope (MPa). B: unit lateral response (MPa/N). C: unit ring traction (MPa/N). D: raw unit chord torsion (MPa/(N mm)). The plate has 25,209 triangles; gray marks the envelope's 75 tip-excluded cells and ring field's 14,875 cells in the producer-masked half-plane, which includes the seat and unloaded blade. The fourth eligible response channel, in-plane tip pry, is shown separately in main Figure 7C. Each field keeps its own physical scale.

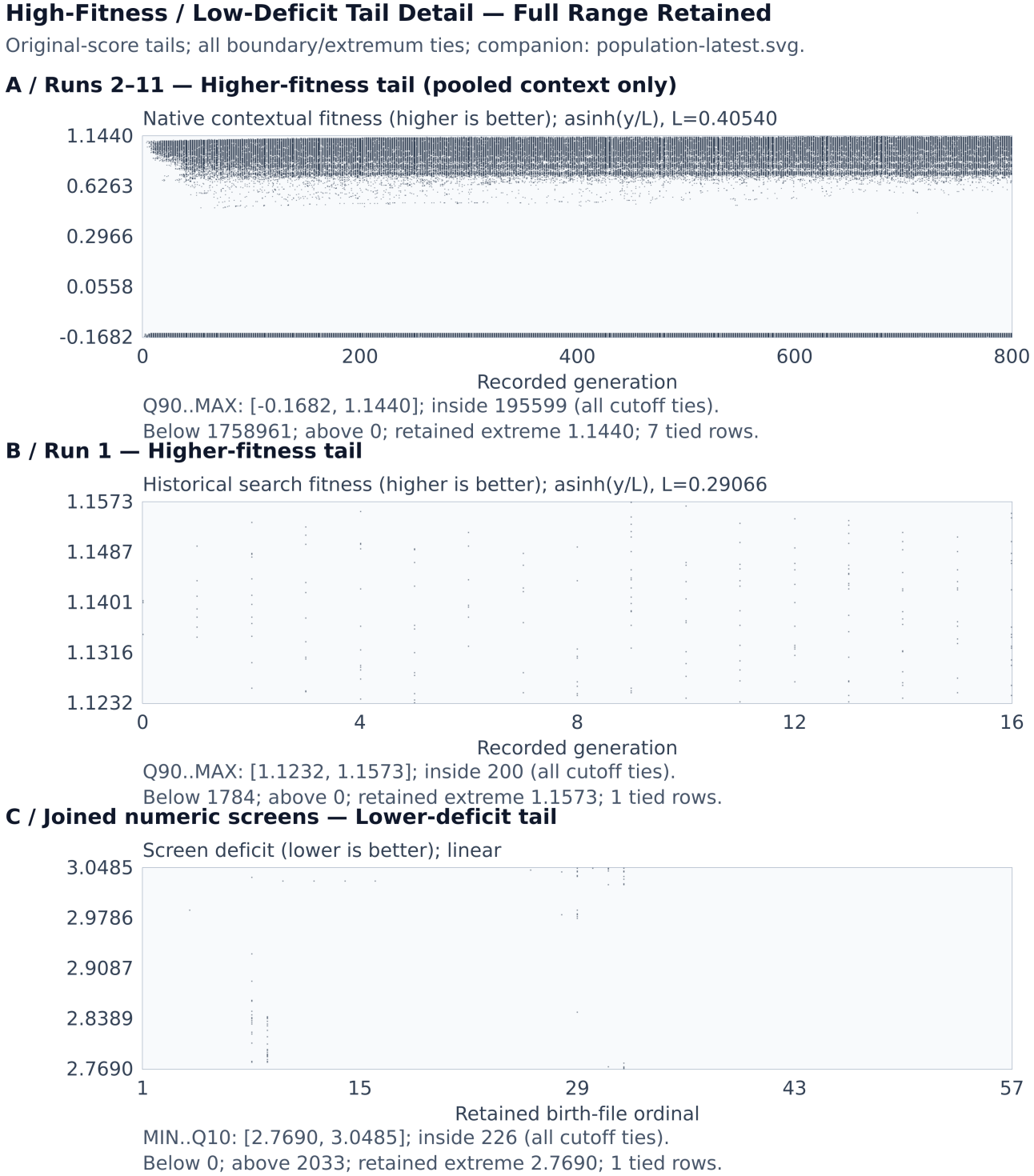


Fig. 6. Directed tail detail of the same retained observations as main Figure 8: native and historical higher-is-better fitness use empirical Q90 through the actual finite maximum; lower-is-better screening deficit uses the actual finite minimum through Q10. Quantile indices are $\lfloor p(n-1) \rfloor$. All cutoff and favorable-extremum ties remain inside, with one point operation per selected observation. Each panel prints its original-unit bounds, transform, exact below/inside/above counts and retained-extremum tie count. The opposite tails remain in the full-range main figure; the 1390 missing screens remain nonnumeric. Marks can overlap; occupancy does not encode density. Pooled native context establishes no comparable global engineering winner. These views add no observations.

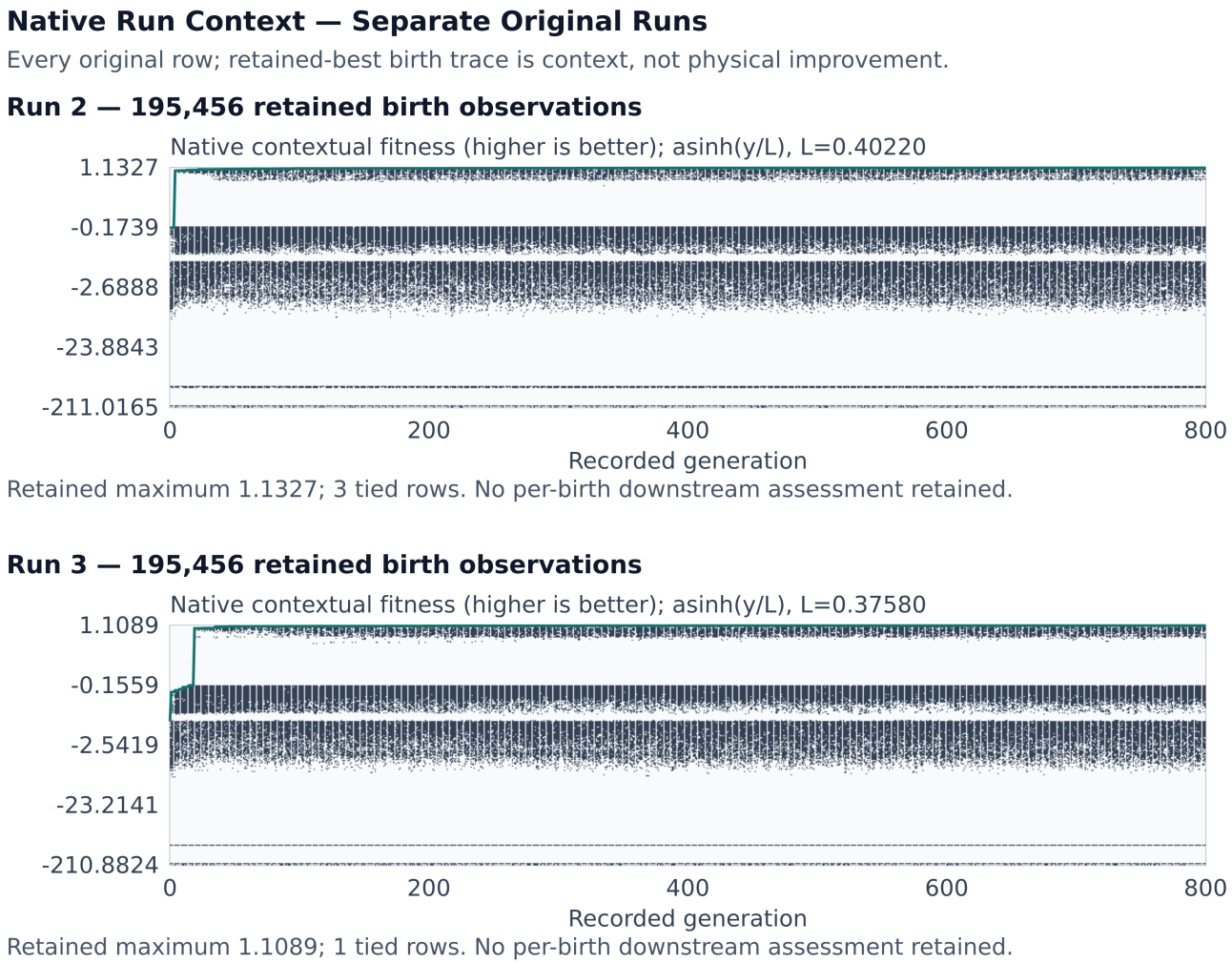


Fig. 7. Retained birth observations and within-run retained-best context for Runs 2–3. Across the five paired views, all 1,954,560 original native births are reused. Each run keeps its full numeric range and printed $\operatorname{asinh}(y/L)$ key; L is that run's empirical 25th percentile of nonzero absolute scores, while zeros remain plotted. The secondary retained-best trace is monotone by construction within its own run and does not connect different runs. It establishes neither physical convergence nor a common scientific quality scale or downstream assessment.

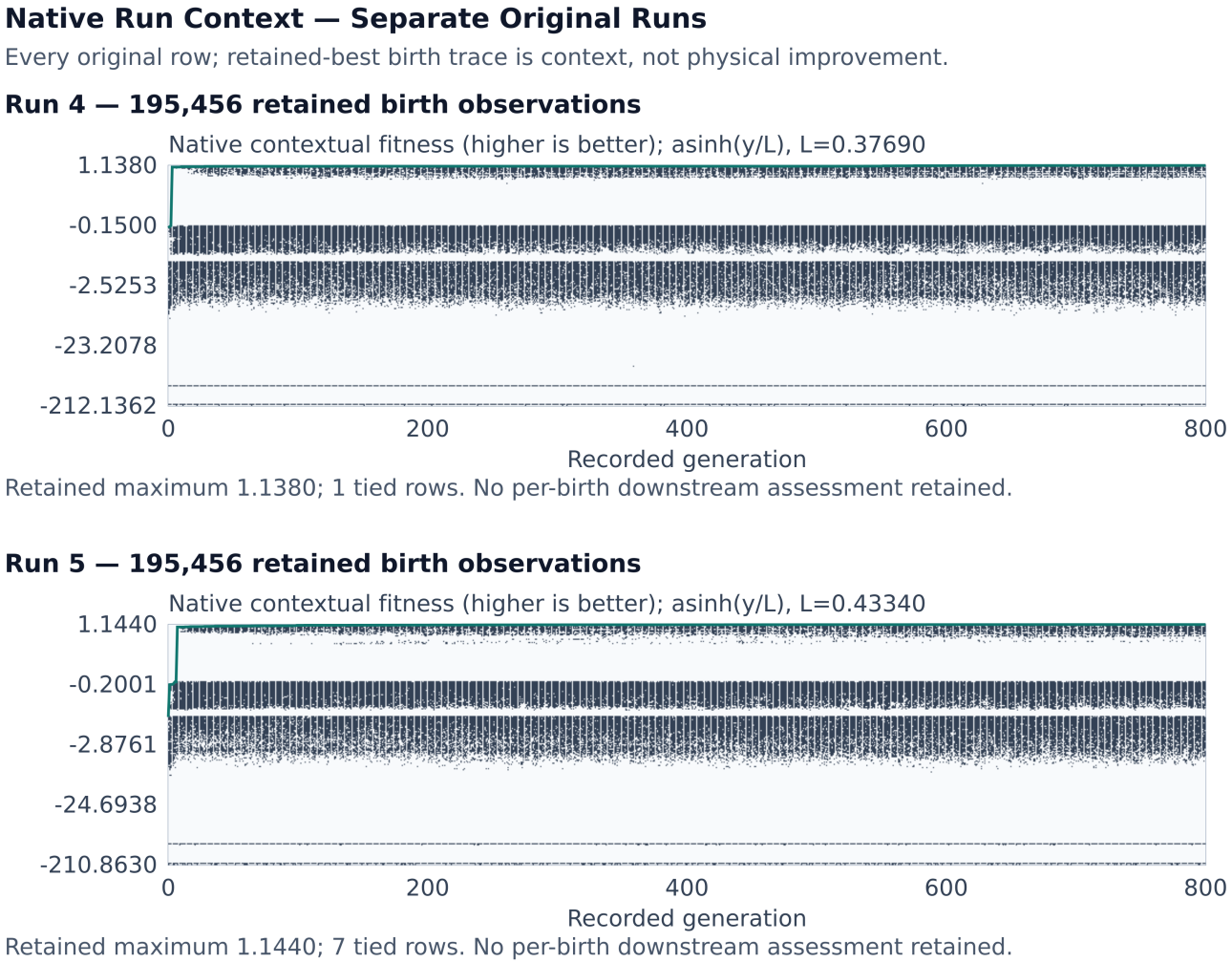


Fig. 8. Retained birth observations and within-run retained-best context for Runs 4–5, using each run's separately printed scale.

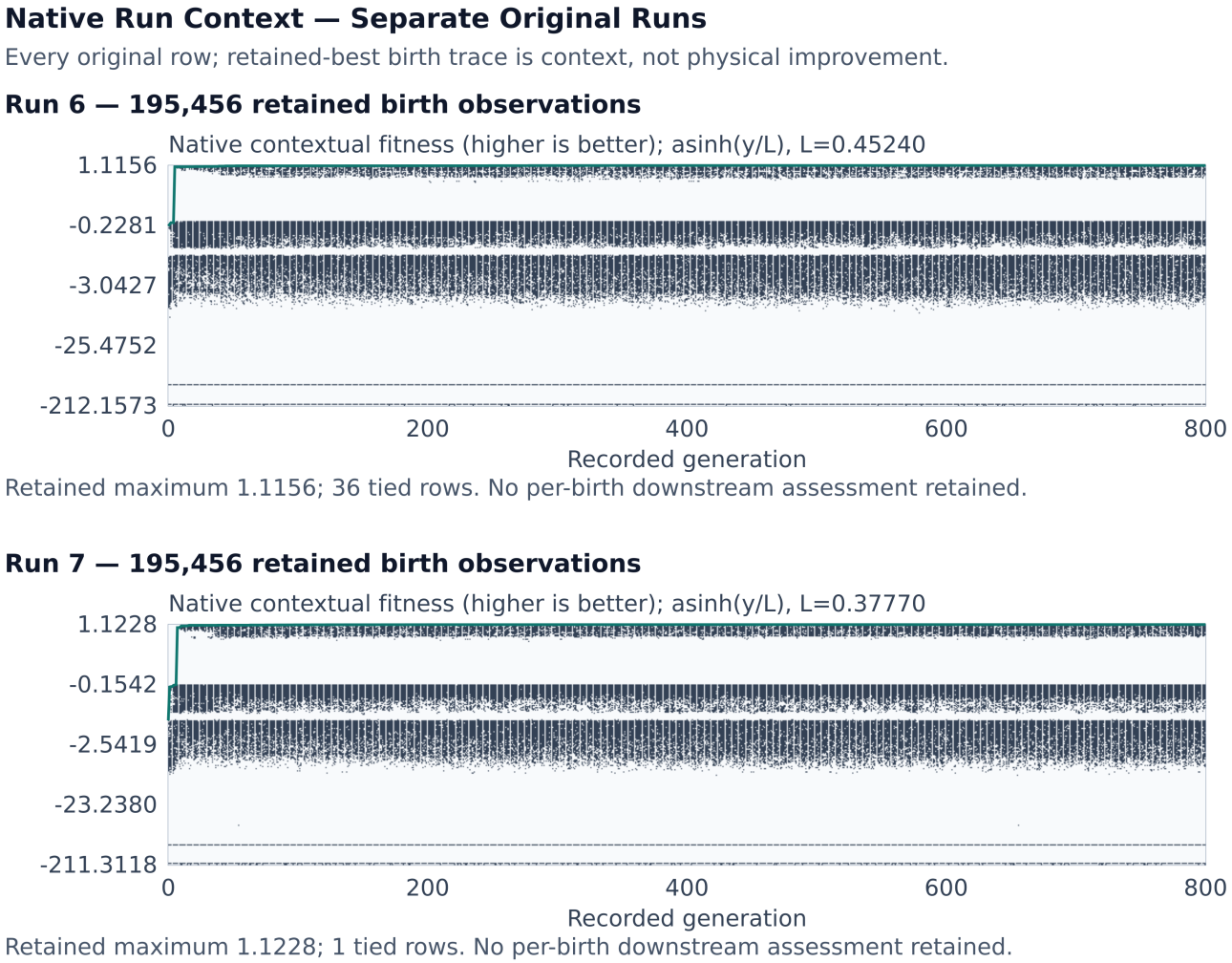


Fig. 9. Retained birth observations and within-run retained-best context for Runs 6–7, using each run's separately printed scale.

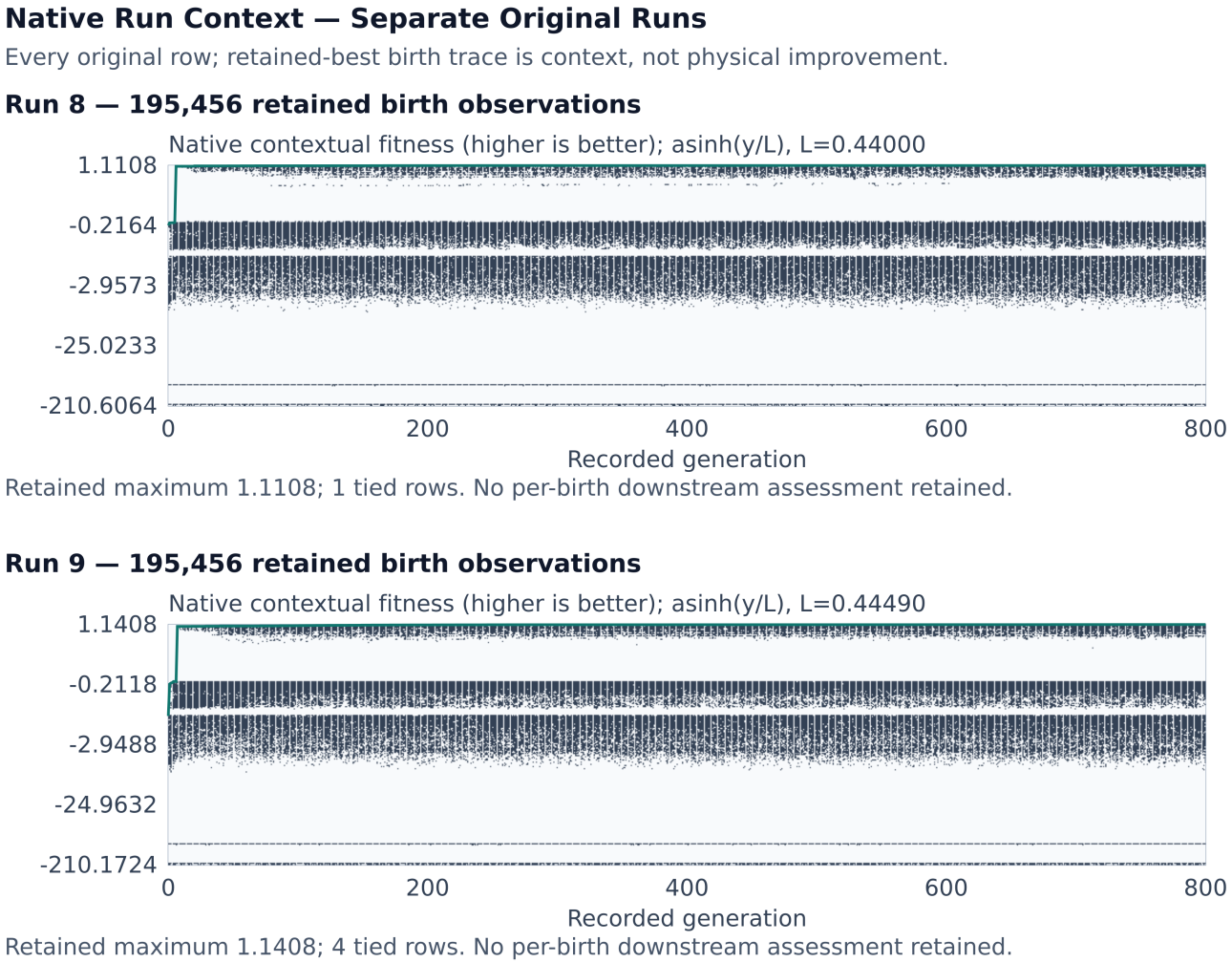


Fig. 10. Retained birth observations and within-run retained-best context for Runs 8–9, using each run's separately printed scale.

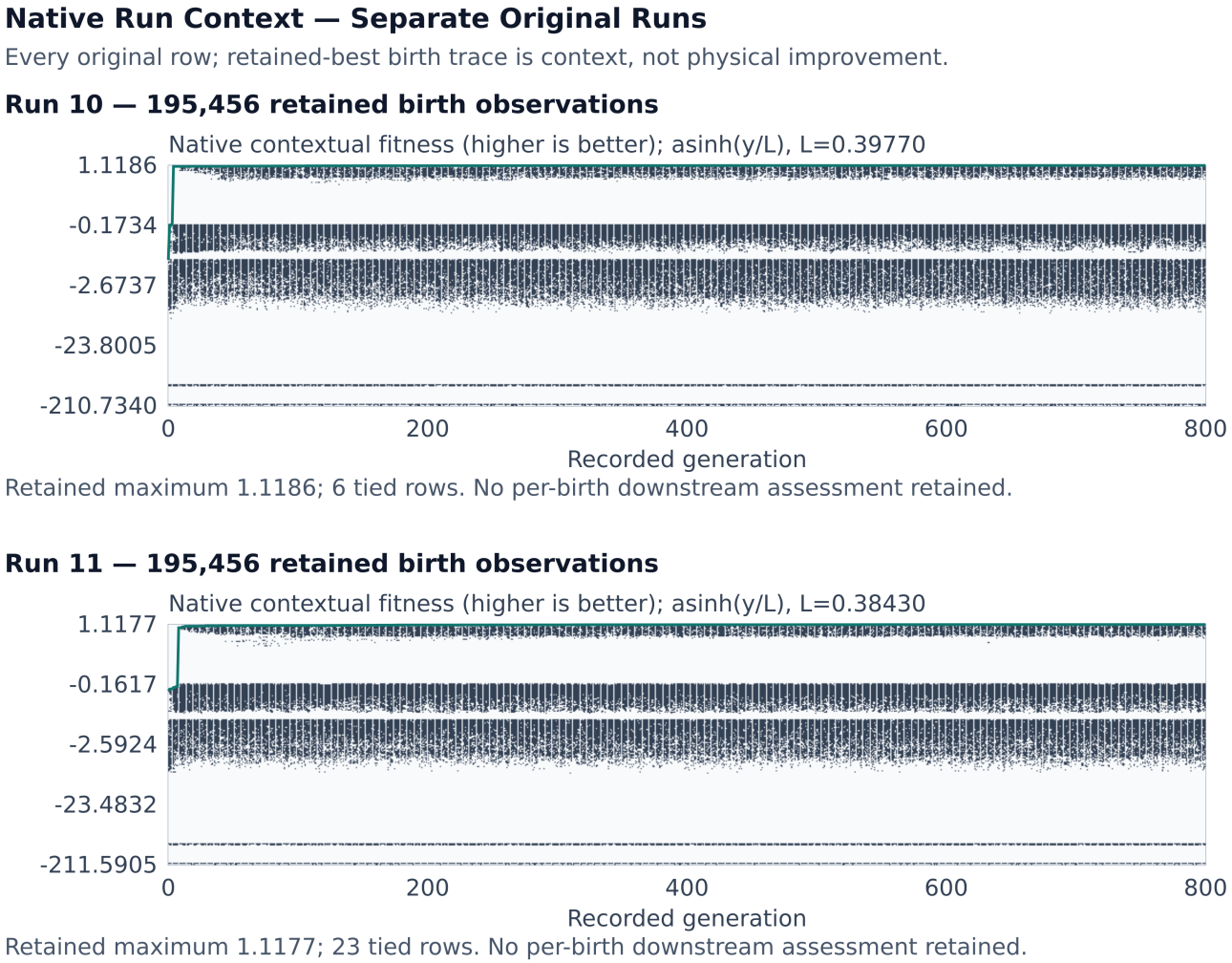


Fig. 11. Retained birth observations and within-run retained-best context for Runs 10–11, using each run's separately printed scale.

D. Bounded Run 5 Display Comparison

For a within-run point view, Figure 12 maps every retained Run 5 birth using $x_+ = g + 1$ and $z = (b - f)/s + \varepsilon$, displayed on base-10 logarithmic axes. Here $b = 1.1440$ is the fixed final best retained score of Run 5, $s = f_{\max} - f_{\min}$ is its full original numeric range and $\varepsilon = 10^{-6}$ is a display floor. The inverse is $f = b - s(z - \varepsilon)$; generation is recovered by $g = x_+ - 1$. Vertical ticks recover original stored fitness, and zero and negative scores remain admitted. The reference is retrospective, not the current running best or an unknown optimum; the floor permits zero original gap and is not a measured residual. Shifted-gap ratios and plotted slopes have no physical performance interpretation.

Figures 13, 14 and 15 reuse the same retained cumulative-best and full-population median summaries. These summary panels draw no raw birth points. Figure 12 separately projects every Run 5 birth and derives its cumulative-best overlay from those raw rows, without a median overlay. The full numeric point universe and its 1390 nonnumeric proposals remain in main Figure 8; none is reassigned a display-floor score. Each comparison uses the complete compatible original Run 5 range, rather than deriving normalization from the summary trace or a crop. Exact parameters, source membership and unchanged leader ties accompany the generated story receipts.

The separate all-point positive-coordinate set keeps twelve source scopes apart: ten native runs, historical births and numeric screening observations, with positive-affine and positive-gap variants for each. The proof index docs/paper/generated/population/positive/index.v1.json binds their source membership, point accounting, display parameters and outputs. This is a set of separate display domains, not a common score calibration. The raw-point variants use a navy–teal–yellow–orange–red key for original stored scores, with a monotone contrast parameter derived from the declared logarithmic coordinate and reversed when needed so that warmer means better within the named run/stage/profile. Original-unit colorbar ticks and the exact contrast definition accompany each view. Lower screening deficit is better; no color scale pools its values with native fitness. Later original-source marks can overpaint earlier colors, including overlapping fixed point footprints; the key encodes the last painted original score, not density or a pixel average. The cumulative-best curve keeps the same coordinates with a dark stroke and light halo. These RGB changes do not rescore any observation. Its full admitted universe remains 1,958,803 numeric observations; 1390 missing screens remain nonnumeric. Screening retains its one-based birth-file ordinal rather than an invented generation or shared campaign trajectory. These views add no observations or physical convergence evidence.

IV. Numerical Methods, Source Bindings and Observation Limits

This supplement interprets retained observations for Individual 2000; it reports no new solve. The original field, assessment and verification input contracts bind the producer revision e8cb049df7d03c7d3113556812cc140ca99a29c3. The producer source and raw-input hashes must be checked against those original contracts. A later figure-semantic ledger describes presentation and cannot independently certify a scientific interpretation.

A. Actual Solid Implementation Chain

The retained runtime capture records image ghcr.io/fenics/dolfinx/dolfinx:v0.9.0 and image identity sha256: 628f40dd62c7f37962ccadc5a745f23d52c56b2ac3ad9b9b6a86c567c07aa27b. The retained comparison sidecar records NumPy 2.5.4 at /ext_sp/numpy/__init__.py. This differs from the current ops/ci/evaluate_cad_candidate.py launcher, which preimports image NumPy and mounts SciPy's package and libraries. This difference further limits reconstruction of the historical environment. The orchestration writes this capture before checking whether valid cached solid outputs can be reused. Neither the original solid record nor the original verification input contract binds the runtime-file digest. These captured versions and image identity must therefore not be promoted to a proven exact execution environment for the historical observations.

The primary code contains a direct-solver fallback, but the retained run explicitly reports the PyAMG/CG branch. The comparison interpolates von Mises stress into DG1 at cell vertices; it is not an L_2 projection, volume weighting or nodal smoothing. Shared incident-cell recovery produces the reported scalar extrema. The primary safety factors are stored to three decimal places, 0.669 and 0.726; the comparison stores four, 0.6693 and 0.7263. Differences computed from these rounded safety-factor tokens are $|0.6693/0.669 - 1| = 0.0004484$ and $|0.7263/0.726 - 1| = 0.0004132$. The code binding and these values support the selected shared-input code-verification comparison. They do not provide tighter unrounded agreement, a global proof of solver correctness or independent validation of the common preprocessed model.

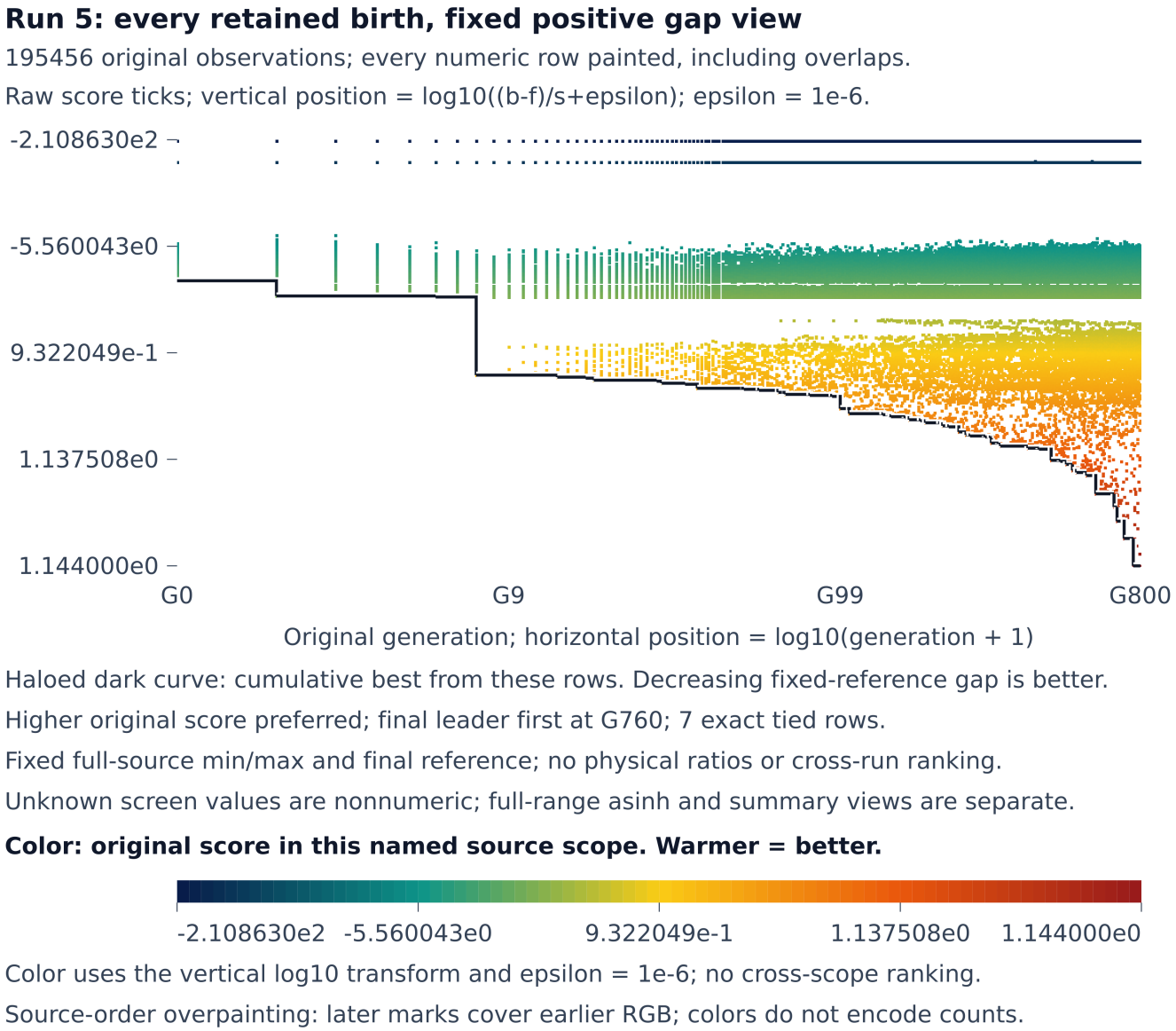


Fig. 12. All 195,456 retained Run 5 births, one point operation per original numeric row. Navy–teal–yellow–orange–red maps original fitness through the declared monotone log-axis contrast; warmer means higher stored fitness within Run 5. Source-order marks can overpaint earlier colors; visible color is not density. The dark cumulative-best curve with a light halo uses those same rows, without a median overlay. Positions retain the fixed full-run range and retrospective retained-best reference; ticks recover original fitness and generations. The earliest of seven exact maximum ties is original ID 185541, public alias 781909, at generation 760 with $f = 1.1440$; no higher stored value appears through generation 800 at serialized precision. This is neither physical convergence nor a global winner or later CAD/solid assessment. Main Figure 8 retains all separate-profile numeric observations and 1390 nonnumeric unknown proposals.

B. Plate Stages, Fields and Regions

The producer's CERT_AREAS are 0.25 and $0.125\,\mathrm{mm}^2$. Certification takes the specified minima over both unbiased solves; its retained spatial fields come from $0.25\,\mathrm{mm}^2$. In the assessment record, lat_rob_n and twist_tipzone_sf therefore denote certification values. strict_lat_rob_0p0625_n and twist_tipzone_sf_strict denote the separate unbiased $0.0625\,\mathrm{mm}^2$ stage. The respective saved pairs are $168.9008053/168.1786057\,\mathrm{N}$ and $0.523920871/0.528698267$.

Lateral and torsion fields use DKT bending stresses recovered as a maximum over three extrapolated corners. Pry and ring fields use T6 centroid von Mises stress. In particular, /mesh/pry has $25{,}209$ stored unit-load values and peak $2.9181\,\mathrm{MPa/N}$. Its producer applies a unit in-plane load perpendicular to the ring-to-tip chord with the ring bore clamped. It must not be labeled as the multi-case envelope in MPa. The original field contract's display-field



Fig. 13. Run 5 summaries on the declared $\operatorname{asinh}(f/L)$ scale, with linear original generation and original-score ticks. The run's printed L belongs to this scope, not the pooled native overview. The retained-best trace is monotone by construction; its final stored-score plateau is not physical convergence or qualification.

list omits this array. Its interpretation here is additive, checked against the bound producer sources; the original contract bytes remain preserved.

The ring producer zeroes cells whose centroid satisfies cen[e].0>=sx0-3.0, a half-plane criterion rather than a seat-only exclusion. The retained ring field has $10{,}334$ active cells and $14{,}875$ source-masked cells. These mask zeros are not measured zero stresses.

The envelope scales lateral/pry fields by $260\,\mathrm{N}$, ring by $1500\,\mathrm{N}$ and raw torsion by $1820\,\mathrm{N\,mm}$ times the producer's global torsion correction. The factor is selected at the peak section and applied globally rather than recovered separately at each location; this approximation does not establish a three-dimensional stress field. It masks 75 tip cells and has peak $1388.6695\,\mathrm{MPa}$. Raw torsion retains its $34.8589\,\mathrm{MPa/(N\,mm)}$ tip-patch peak; the post-$4\,\mathrm{mm}$ exclusion capacity peak is $0.6611\,\mathrm{MPa/(N\,mm)}$. Body torsion searches beyond $15\,\mathrm{mm}$, while the tip-annulus observable searches 4–$15\,\mathrm{mm}$. No strict spatial field or full volumetric solid stress array is retained in the publication inputs.

C. Three-Level Study and Refusal Semantics

The independent plate study has maximum areas 4, 1, $0.25\,\mathrm{mm}^2$, with nominal $h = \sqrt{A_{\max}}$ and refinement ratio 2. This convention uses the requested maximum area, not the measured distribution of realized element sizes. In fine-to-coarse order, the yield-order proxy is 1.4381383, 1.5763238, 1.5948326 and the lateral safety factor is 0.6535320, 0.6886658, 0.6810133. The stored relative-GCI procedure records valid sequence/observable metadata but refuses the former for nonpositive observed order ($p = -2.9003199$) and the latter for oscillation. Extrapolated values and fine-grid estimates are null. A valid sequence descriptor is not an admitted extrapolation, and refusal is not a universal statement about other error-estimation methods.

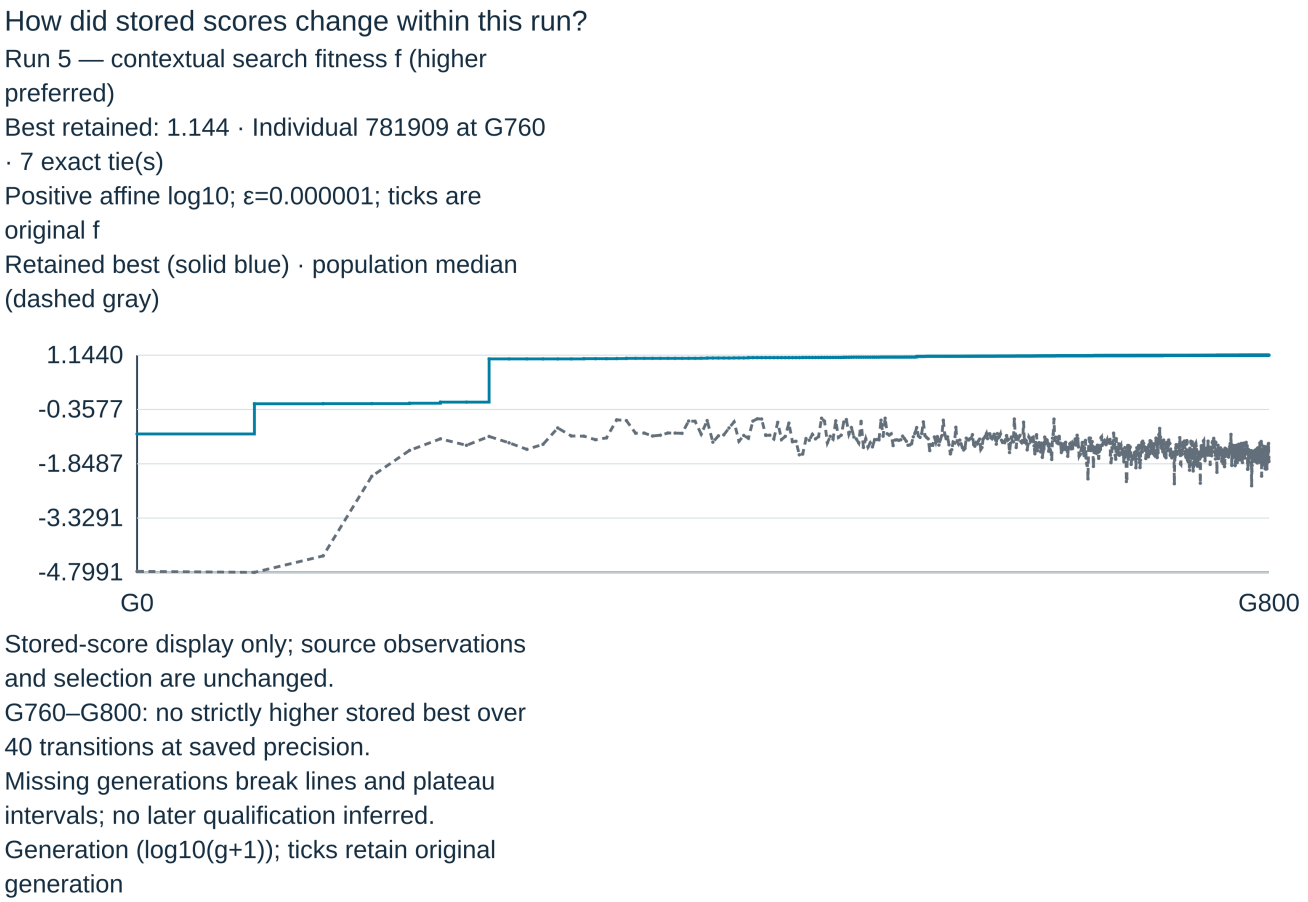


Fig. 14. The same Run 5 summaries with positive-affine logarithmic display: $x_+ = g + 1$, $y_+ = (f - m)/s + \varepsilon$, where $m = f_{\min}$, $s = f_{\max} - f_{\min}$ use the full original compatible run and $\varepsilon = 10^{-6}$ is a display floor. Original-score ticks use $f = m + s(y_+ - \varepsilon)$; generation ticks recover $g = x_+ - 1$. The summary gap view in Figure 15 labels normalized gap; the all-point main gap view instead recovers original fitness ticks. These summary panels draw no raw birth points and change neither the source domain nor saved summaries. Shifted-score ratios and log slopes have no physical interpretation.

D. Other Observation Families

Representation/export records bind genome, material, B-rep, surface mesh, mass and their digests. For Individual 2000, the saved export reports one solid and one shell, a valid OpenCascade B-rep and a $44{,}994$-vertex/$90{,}000$-triangle display mesh. The reference comparison uses a 4230-point exterior contour and 1117-point authored reference; saved distance $2.285\,\mathrm{mm}$, curvature EMD 0.0071 and similarity 0.4079 describe conformity to that prior. Export metrology and nominal coordinates are distinct observations; neither establishes comfort or human aesthetic approval.

The regional sequence proxy takes the governing ratio of regional elastic capacities over lateral, torsion and pry cases, with a stored target of 1.2. It does not model fracture, crack initiation or plastic redistribution. The modified-Goodman scalar obeys $n_f^{-1} = \sigma_a/S_e + \sigma_m/S_u$ with $\sigma_a = \sigma_m = \sigma_{\max}/2$ for the modeled zero-to-maximum cycle; its saved value is 0.47179485. No spatial fatigue array or measured life is supplied. The source cyclic lateral reference is $180\,\mathrm{N}$ with a zero-to-maximum modeled cycle. The overload scalars are capacity divided by the producer's reference load: $16{,}200\,\mathrm{N\,mm}$ torsion, $1000\,\mathrm{N}$ in-plane tip, $3427\,\mathrm{N}$ spine and $3000\,\mathrm{N}$ ring-back (twice the $1500\,\mathrm{N}$ pull reference). Larger factors mean greater modeled margin and unity denotes the named-load reference. The in-plane tip factor is record-only under the bound historical policy. These factors differ from the local tip-torsion factor normalized by $1820\,\mathrm{N\,mm}$ and do not establish overall qualification.

The alternate support study records radial, tangential and flange springs of $50{,}000$, 5000 and $100{,}000\,\mathrm{N/mm}$ and a Winkler grip foundation of $200\,\mathrm{N/mm^3}$. Its retained static lateral ratio is 5.2026756, conditioned on the

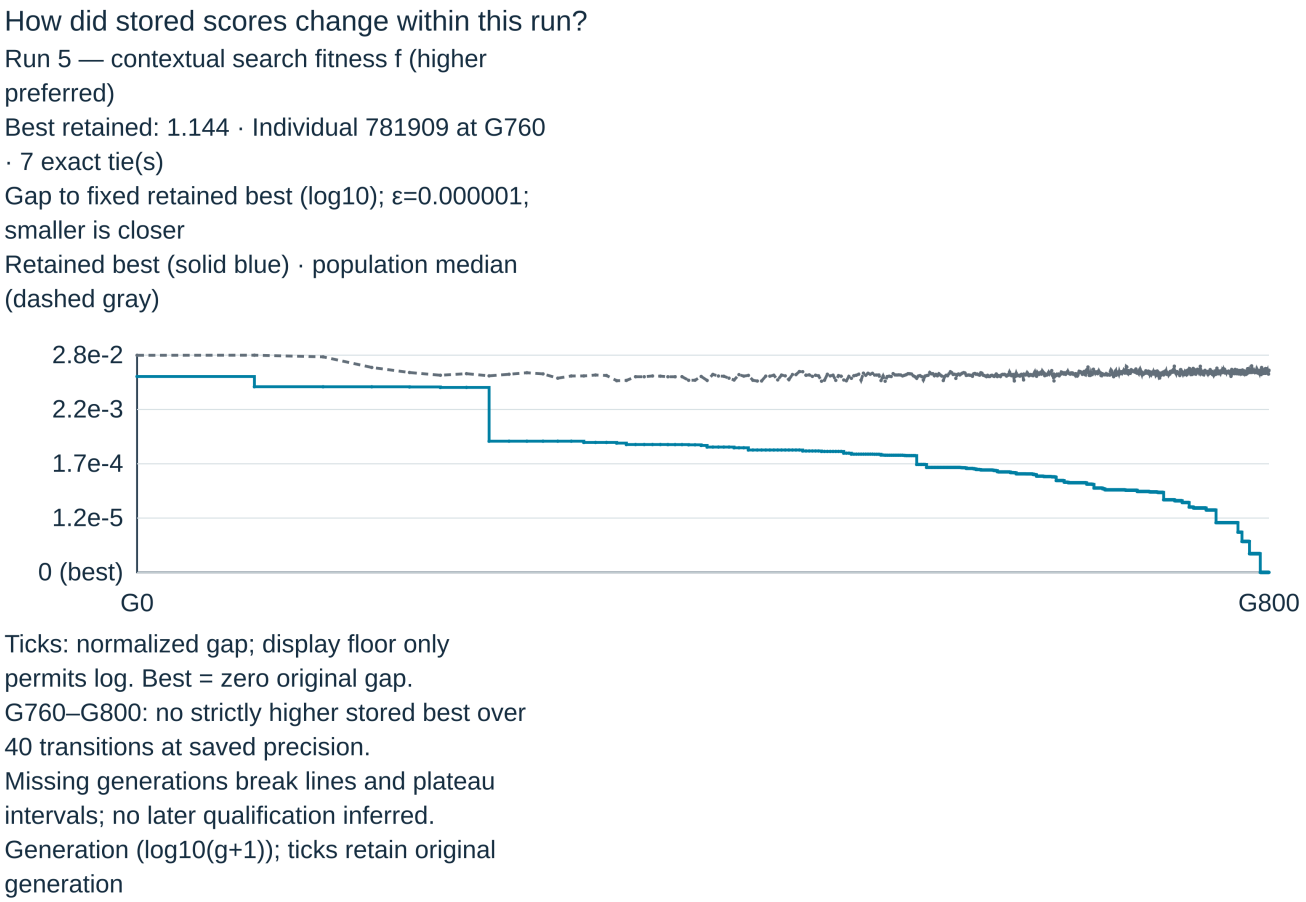


Fig. 15. Run 5 retained cumulative-best and population-median summaries in the positive-gap display, with no raw birth points. The fixed own-run final reference, full original range, display floor and inverse match the definitions in the main text. Here ticks show normalized gap, with zero original gap at the retained best, and original generations; the separate all-point main Figure 12 uses original-fitness ticks. The seven exact maximum ties, earliest retained attainment at generation 760 and final 40-transition stored-score plateau are source observations, not physical convergence, qualification or a global winner.

alternate supports rather than replacing the certification ratio 1.4353236. The original record also stores eigenvalue 7.0350619 and status pass; both are preserved here solely as historical provenance. The producer applies a symmetric eigensolver to $\mathbf{K}_s^{-1}\mathbf{G}$, but the required operator self-adjointness and eigenpair residual are unvalidated in the retained evidence. This eigenvalue is not admitted as a current engineering buckling result, and any qualification inference depending on it remains unestablished. Original scalar/status bytes and historical score records remain unchanged; no replacement solve, new eigenvalue, rescore or synthetic mode shape is supplied.

Finally, the historical battery has 23 subjects and 18 columns (414 status cells), including shared software controls, side-study references, blocked geometry bindings and 20 solid-file completeness labels. The original 44-entry crosswalk and historical matrix remain preserved by the publication package. Definition counts (178 assurance challenges/356 paired slots) and repeated control labels are separate from actual execution receipts and from candidate-specific mechanical evidence. A favorable completeness flag is not a favorable strength factor.

V. Cohort Inventory, Overlap and Descriptive Summaries

The inventory separates original run identity, retained capture and record kind. Receipt observation times below identify accession observations, not independently verified historical OS process start/stop times. Exact raw fields and source precision remain in the immutable JSON accessions.

TABLE IX

RETAINED SOLID IMPLEMENTATION CHAIN. RETAINED ENVIRONMENT CAPTURES ARE DISTINCT FROM PROVEN HISTORICAL EXECUTION BINDINGS. RUST IS THE PIPELINE/ORCHESTRATION LAYER, NOT EITHER SOLID ASSEMBLY LIBRARY.

Property	**Primary**	**Comparison**
Producer	fem_solid/tip_solid.py	fem_solid/cross_check.py and shared DOLFINx seams
Assembly/field	scikit-fem ElementVector(ElementTetP2), integration order 2	DOLFINx vector Lagrange P_2 on linear tetrahedral geometry
Linear solution	PyAMG smoothed aggregation, CG; recorded algorithm skfem_p2_amg_cg	PETSc CG/GAMG; Chebyshev/Jacobi multigrid-level options
Retained versions	Runtime capture: scikit-fem 12.0.2; PyAMG 5.3.0; Python 3.12.3; NumPy 2.5.4; SciPy 1.18.1	Sidecar: DOLFINx 0.9.0; PETSc/petsc4py 3.22.0; Basix 0.9.0; UFL 2024.2.0
Convergence record	152 iterations; relative residual 8.803574×10^{-11}	158 iterations; positive KSP reason 2; sidecar residual norm 1.324748×10^{-10} (a different reported norm)
Common mesh	Saved NPZ with $91{,}008$ tetrahedra and $416{,}583$ displacement degrees of freedom; mesh-file SHA-256 d3f7a444175fb6a6f64ee1118a00f2e83d6994c66b1c266cfd94ac14572faab1. This file identity differs from the sidecar's internal mesh-layout hash.	
Material/load	Ti-6Al-4V isotropic elasticity, $E = 113.8\,\mathrm{GPa}$, $\nu = 0.34$; common saved allowable and $1820\,\mathrm{N\,mm}$ chord torque. Saved primary and comparison scalar outputs use different rounding.	
Boundary	Zero displacement on cut-arc vertices and midpoint degrees of freedom on edges with both ends fixed. Minimum-norm z-forces at vertices within $2\,\mathrm{mm}$ of the tip impose torque with zero resultant force and zero transverse moment. No transferred-displacement boundary is recorded.	
Recovery/region	Element-vertex von Mises stress, then incident-cell vertex maximum or unweighted mean; planar tip distance in $[4, 15)\,\mathrm{mm}$ and nearest saved cut-vertex distance $> 3\,\mathrm{mm}$. The comparison sidecar records 6930 ROI vertices.	

A. Coverage Sources and Status Qualifications

H: docs/paper/JR_EI_P1_T04_HISTORICAL_RECONCILIATION_RECEIPT_20261019.json reconciles 1984 births, 2176 memberships, 19 event frames and selected later strict records for historical Run 1. Its 146-column CSV retains numeric coordinates/material; its event stream supplies recorded within-run parentage, while external seed parent identifiers are unrecorded. Selected geometry/fields do not imply per-birth CAD coverage.

P: docs/paper/JR_EI_P1_T05_NEWER_BIRTHS_SCREENS_JOIN_RECEIPT_20261019.json admits 57 proposal/screen file pairs across 24 directories: 3649 proposals, 2259 one-based joins and 1390 missing screens. The saved screens lack the later ultralight weighted fitness and remain a separate profile. Proposal records retain certificate identity, family and genome row, but not per-child parent identifiers. Run-level seed metadata does not supply those missing parent links. Selected exports and summaries do not establish all-proposal downstream assessment.

N: docs/paper/JR_EI_P1_T08_LOSSLESS_GEN800_POPULATION_ACCESSION_RECEIPT_20261019.json admits ten complete 146-column population tables (128 numeric coordinates plus material within 146 total fields), 1,954,560 rows and 200 compressed segments. The original CSVs total 3,811,896,749 bytes; compressed segments total 494,843,141 bytes. The admission receipt records lossless identity. Parent columns are absent from these tables; other retained event streams do not by themselves establish a complete analyzed genealogy. Twenty-four separately retained STL artifacts are inventoried apart from the population payload and do not establish per-birth geometry binding. Later CAD/solid assessments and spatial arrays are not retained per birth in this accession. The population index records d108c9d7d8149ac230ca13fb642159985a8f2857_with_untracked_worktree_state_per_T01_V2 as its source-state provenance, with no process-configuration hash for five of the ten completed runs. P1-T06 separately records baseline 009f5ac520fa3afe8ed8025230eef604dbb7f922, local-patch SHA-256 fb80e34dfde1f3b49788e1d71e6700ef3da25f11e5270ed0b5dcab79e083b219 and patched qd.rs SHA-256 dfce6905486e51c60c0e60ae307be881ab8a0c58fb7b7d7fdba3a0841b69103f. Those differing provenance records do not establish a clean executed source tree or historical process binary.

E: docs/paper/data/selection_receipts.v1.json binds the editorially illustrated Individual 2000 to the automatic

TABLE X
EARLIER SUMMARY CAPTURES AND OVERLAP WITH COMPLETED RUNS. EXACT PACKAGE IDS ARE RETAINED FOR CROSS-CAPTURE IDENTITY; CAPTURE HASHES ARE IN THE ADMITTED P1-T06/T08 RECEIPTS.

Original package ID	Earlier rows	Generation range	Later completed run
evo_full_20261018_xbabe2_c01	1900	0–474	Not in completed set
evo_full_20261018_xbabe2_c02	64	0–15	Not in completed set
evo_full_20261018_xbabe2_c03	72	0–17	Not in completed set
evo_full_20261018_xbabe2_c04	16	0–3	Not in completed set
evo_full_20261018_xbabe2_c05	2132	0–532	Not in completed set
evo_full_20261018_xbabe2_c06	908	0–226	Not in completed set
evo_full_20261018_xbabe3_c01	2836	0–708	Run 6
evo_full_20261018_xbabe3_c02	2792	0–697	Run 7
evo_full_20261018_xbabe3_c03	2748	0–686	Run 8
evo_full_20261018_xbabe3_c04	2792	0–697	Run 9
evo_full_20261018_xbabe3_c05	1412	0–352	Not in completed set

manifest and its original field/assessment/verification contracts. This stable publication alias has original registry origin RetainedExport; it is not a rank, generation or evidence of the 2000th search birth. An additive content join links it to the second proposal in evidence/evolve_certifiable/automatic/round_00_births.json (zero-based /genomes/1, round 0, seed 20261017). Both its certificate db5d8e32e99d0b2d8ef6d7c7f9622a0601487c25b80879bbc8be1a502693c55e and genome CSV SHA-256 a526b2ef73b25e51a56bcfab1f67640bcbe734955b95a6b432a99dab5cd9f654 match. The genome bytes are reconstructed as the retained header, LF, row and LF. This recovers a proposal identity, not actual parent identifiers; numerical similarity to another genome is not parentage evidence. CAD and plate fields are available; full solid, strict plate, eigenvector and fatigue spatial arrays are not publication inputs. This record is separate from the historical 23-subject battery. C: the controlled-probe CAD scorecards bind Individuals 2001 and 2007; their ancestry to a particular evolutionary birth remains unestablished. In all rows, physical observations are not reported in this study. No absence is strengthened to "not performed" without evidence of that fact.

B. Current Native Directories and Earlier Summary Captures

P1-T06 observes its native-summary accession at 2026-10-19T17:41:30Z. P1-T08 records its population accession at 2026-10-19T18:22:30Z. Those are receipt times. The current directory universe has 18 members: ten completed generation-800 directories and eight others. The earlier eleven-file snapshot contains 17,672 summary rows, reaching generation 708. Four original identities occur in both captures. Public Runs 6–9 name those completed runs; they are not four additional replicates merely because earlier summary captures exist.

The eight other current directories are xbabe1/c02, xbabe2/c01--c06 and xbabe3/c05; their retained summaries have differing endpoints. The nine aborted crescent attempts are separately admitted by P1-T03: seven retain 216 live-history rows in total and two retain no CSV rows. Aborted island attempts and their proposal files are another record universe; they are not added to these nine crescent attempts or the completed-population denominator. A complete retained table or capture does not establish full-state restartability: native disk-only fresh-process resumption remains unproven. Neutral integer-stream recovery fixtures have their own state and execution receipts and are not evidence of resumed native searches.

C. Ten-Run Descriptive Summary

The run is the unit for these descriptive comparisons. Sample standard deviations use $s^2 = \sum_{r=1}^{10}(x_r - \bar{x})^2/9$; no confidence interval or independence of births is asserted. Native fitness includes contextual novelty, so maxima summarize retained preference under each run's context rather than a common physical-performance measurement.

The six weighted-reporting files contain fifteen summary rows; those files alone do not retain a full underlying population or CAD bundle. For Individual 2002, another accession does retain three round birth/screen pairs under evidence/evolution_waves/2026-10-17/xbabe3/out/w411c3/. Its content matches the twelfth proposal in round_01_births.json (zero-based /genomes/11, round 1, seed 3412): certificate ee2ec73480f12773579cf573ac3cc0

TABLE XI
COMPLETED NATIVE RUNS. EACH HAS 195,456 BIRTHS AND 3204 GENERATION/ISLAND SUMMARY ROWS. THE ORIGINAL RECORDS RETAIN ADDITIONAL DISTRIBUTIONS AND EXACT PRECISION.

Run	Search-legal births	Maximum stored fitness	Generation-800 legal median
Run 2	74874	1.1327	-0.2462
Run 3	77011	1.1089	-0.2211
Run 4	77396	1.1380	-0.1933
Run 5	77625	1.1440	-0.2225
Run 6	70278	1.1156	-0.2534
Run 7	74777	1.1228	-0.2023
Run 8	71550	1.1108	-0.2351
Run 9	76284	1.1408	-0.2072
Run 10	74309	1.1186	-0.2302
Run 11	80132	1.1177	-0.2610

838734e8caf40411e6bcba0bb12cf1210b and exact header/LF/row/LF genome SHA-256 078c27c765791cb8b9f8ce29 f2047ea0a8b0f2f76f7a542643f93643ca95b96e. The corresponding screen has the same certificate and the earlier pending-assessment label. Per-child parents are unrecorded.

The separate evidence/evolution_waves/2026-10-18_ultralight_v1/xbabe3/w411c3/leaderboard.json retains its later-assessment summary: $W_{\mathrm{CAD}} = -7.414831$, $G_1 = 0.259798625$, solid-tip factor 0.3636, 15 battery entries passing and qualified=false; the leaderboard has six assessed rows and zero qualified. The referenced exported CAD and full scorecard paths are not retained in the checked repository tree. A retained summary is distinct from a complete retained underlying artifact bundle. Individual 2002's original registry category and reporting vector remain separate from Individual 2000's geometry and fields. Additive links correct the interpretation without overwriting historical registry or evidence bytes.

VI. Related-Work Interpretation Notes

The main comparison concerns representation, search, evidence retention, and validation scope. Topology-optimization context [13] establishes that computational changes to structural form are longstanding; no implementation equivalence to every method in that review is asserted. DeepCAD and Text2CAD concern CAD sequences and learned generation [2], [3]. The ST5 preprint supplies a concrete earlier evolutionary-engineering example with fabricated prototypes; its report-specific qualification status is preserved [4]. MAP-Elites supplies archive context [5]; actual descriptors and retained coverage must still be reported for Jarambit's cohorts. FunSearch is limited here to proposal/evaluation separation [6].

PROV-DM's entity/activity/derivation/collection distinctions provide established terms for discussing provenance [14]. The present study does not implement or test a PROV interchange profile. FAIR's discovery and reuse goals and software-citation specificity help separate an accessible research object from a checksum alone [15], [16]. A source snapshot is cited as a snapshot; its identity is not substituted for an unrecorded execution environment or for this manuscript's eventual release identity.

Software methods references support library roles, not a numerical model's validation [8], [9]. The cited DOLFINx v1 preprint from 2023 remains a valid publication object; it differs from the retained DOLFINx 0.9.0 environment. The consulted SciPy API states the symmetric/Hermitian operator contract [11]; its current documentation does not identify the exact historical executed environment. The retained SciPy 1.18.1 capture and its cache/execution-binding limitation are described in Section IV. Roache's grid-refinement reference is retained [12]; the bound reporting procedure and its actual refusal rules, rather than the citation alone, establish which estimates were admitted.

Crameri *et al.* motivate attention to perceptual color gradients and accessibility [17]. This citation does not validate Jarambit's percentile composite, lighting, saliency choices, or final figure accessibility. Those require their own definitions and final visual inspection. The NIST artifact supports distinguishing digital geometry from measured process behavior [10]. NASA's standard supplies credibility context; compliance is not asserted [7].

VII. Complete Historical 23-Subject, 18-Evaluator CAD Battery

A. A Matrix of Different Evidence Kinds

The historical battery contains 23 retained subjects and 18 evaluator columns. Its 414 cells include software capability, geometry binding, numerical findings, shared reports, and record completeness. The manifest fixes membership. A later automatic assessment is a separate corpus and does not silently create a twenty-fourth subject. All cells in this historical corpus report fresh_this_lane:false.

The stored statuses reconcile to 221 passes (221), 81 failures (81), and 112 blocked records (112). These are labels from heterogeneous producers. They are preserved here as historical data alongside the exact explanation of what each label meant. In particular, the four shared build columns account for 92 stored passes (4×23), and the local-solid column's 20 favorable statuses concern record completeness. Neither fact supports an aggregate claim of 221 independently successful physical engineering tests.

Figure 16 retains every subject/evaluator pairing in manifest order. Each subsection below refers to the column with its exact evaluator ID and docs/tests/README.md anchor. Letters accompany the colors so missing evidence remains distinguishable from an unfavorable observation. Source-cell selectors, original kinds, and evidence-path strings remain in the linked projection (docs/paper/generated/battery-matrix.json); the aggregate table remains in the artifact package. The manifest's source tag is not uniform across its members: 21 subject records cite 6d18ef8c, while the two w283 records cite 4e54019d. These are retained provenance fields, not a newly verified homogeneous solver regime.

B. Exhaustive Column-by-Column Historical Battery Reference (18 Evaluators)

1) build.analytic: Analytic Software Controls (battery-build.analytic): This column references shared analytic/reference software controls (invariance.rs). Such controls can expose errors in units, invariance, assembly, or solver behavior on reference problems. A per-subject copy of a shared control status does not establish another execution. The historical 23 passes should therefore be understood as repeated references to capability evidence. They do not establish candidate-specific geometry validity or physical behavior.

2) build.gci_contracts: Convergence Arithmetic (battery-build.gci_contracts): This column concerns software rules for interpreting mesh-study data (test_gci.py). Positive and negative controls demonstrate that a formula or refusal condition is implemented as intended. It does not mean that a particular CAD has a suitable three-level study, matching observables, or an admitted asymptotic error estimate. Candidate-specific convergence belongs to plate.gci.

3) build.claims: Reporting and Admission (battery-build.claims): This column references controls over how evidence is accepted, refused, and rendered into claims (check_claims.py). Its object is the reporting system. A favorable software outcome establishes behavior of that control under its fixture; it does not establish the mechanics of any CAD or authorize an unscoped public safety statement.

4) build.dolfinx: Separate-Solver Capability (battery-build.dolfinx): The retained column references shared solver-comparison evidence (evidence/cad_battery/dolfinx_cross_check.json). It indicates the scope of that shared capability record, not a fresh dual-solver computation for every subject. Candidate-specific comparison requires the two actual execution identities, matched problem specification, and observable correspondence (G16-16).

5) geom.unit: Export Units (battery-geom.unit): The unit column (verify_genome_export.py) asks whether the exported representation's unit interpretation is established by the relevant evidence. All 23 historical cells are blocked. This does not prove that every export has incorrect units; it means the required binding was not established in those historical records.

6) geom.one_solid: Topology Binding (battery-geom.one_solid): This column (verify_genome_export.py) records whether single-solid topology was established for the subject. Its 23 blocked cells preserve a missing-evidence condition rather than 23 unfavorable topology measurements. A mesh closure check and a B-rep solid check do not establish the same fact.

7) geom.mass: Mass Observation (battery-geom.mass): The mass column (battery.rs) has 19 retained genome observations and four blocked reference entries. For those genome entries, the historical value is a plate/certification mass observation, not a demonstrated STEP-versus-raster consistency comparison.

8) geom.pin: Geometry Identity (battery-geom.pin): This column (cad_ingest.rs) concerns binding the observation to the intended CAD via geometry_sha256. All 23 historical entries are blocked because filenames alone cannot establish cryptographic identity.

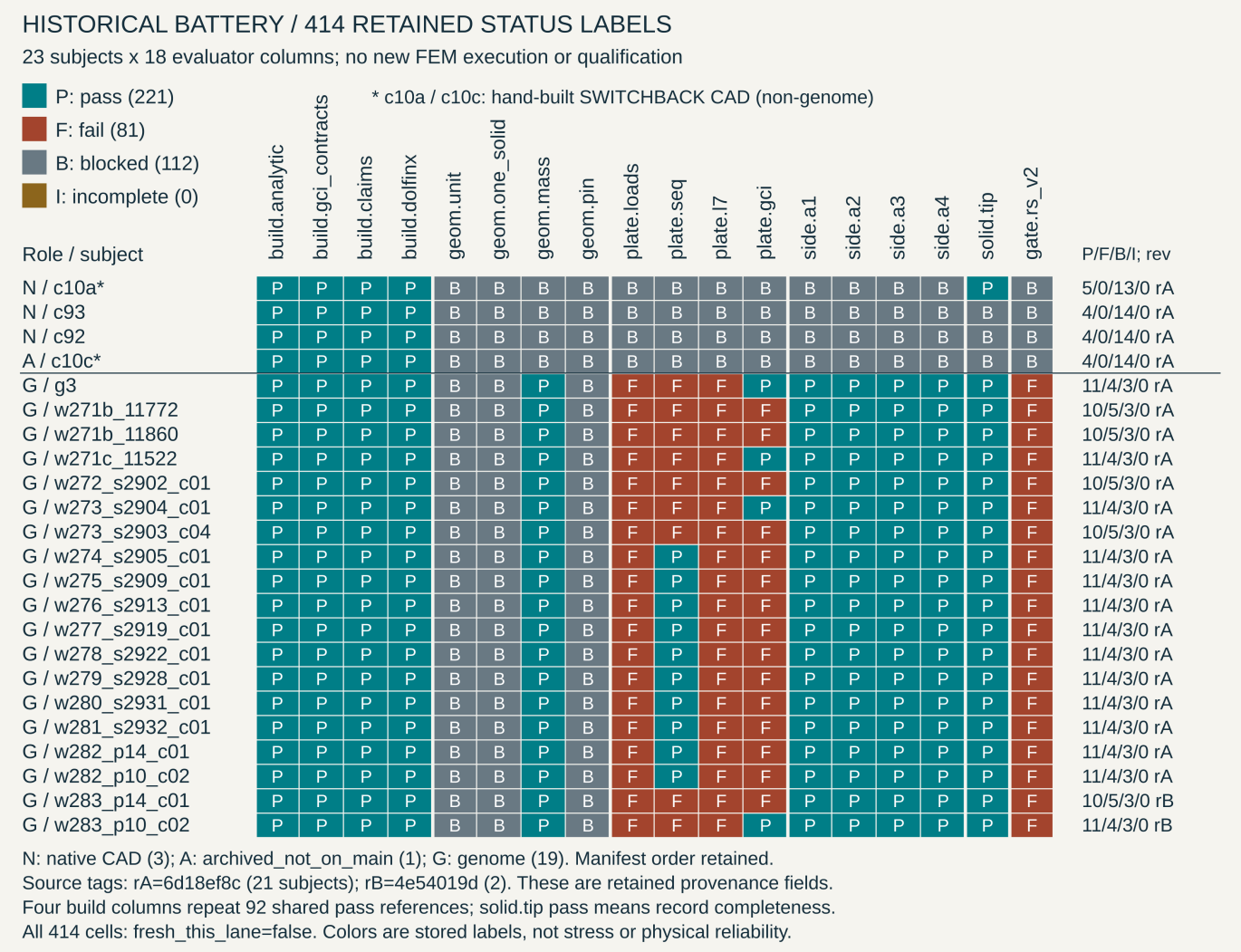


Fig. 16. Supplement Figure S1: All 414 original labels in the fixed historical battery, separate from the cover's assessed example (cert_pipeline_r01_p14_c02). N, A, and G reproduce source roles: native CAD, archived-not-on-main, and genome. P/F/B/I mean stored pass/fail/blocked/incomplete, not common physical outcomes. Source tags rA/rB identify the different retained revision fields (6d18ef8c vs. 4e54019d). Four build columns repeat 92 shared pass references; solid.tip passes mean record completeness. No cell is fresh in this lane, no missing observation is imputed, and no row is promoted to qualification.

9) plate.loads: Plate Response Finding (battery-plate.loads): The load column (fem.rs) reports the stored plate-response comparison: 19 unfavorable records and 4 blocked reference entries. Unfavorable findings are results of the model and comparison policy, not solver crashes; blocked entries are missing records, not measured failures.

10) plate.seq: Regional Response Comparison (battery-plate.seq): The sequence column (fem.rs) records 10 stored favorable and 9 unfavorable findings among genome entries (plus 4 blocked reference entries), inheriting the linear-elastic proxy limitations of G16-14.

11) plate.l7: Fatigue Screen (battery-plate.l7): The fatigue column (fem.rs) records 19 unfavorable model assessments and 4 blocked entries. It is a disclosed analytical screen, not a physical lifetime test.

12) plate.gci: Candidate Mesh Study (battery-plate.gci): This column (gci.py) reports candidate-specific convergence evidence: 4 favorable, 15 unfavorable, and 4 blocked records, distinct from the software arithmetic controls in build.gci_contracts.

13) side.a1: Shared Side-Report Reference (battery-side.a1): References a shared retained side report (g3_side_report.rs): 19 favorable references and 4 blocked entries, representing report availability under the historical producer's convention rather than a distinct executed experiment.

14) side.a2: Shared Side-Report Reference (battery-side.a2): Similarly references the common side report (19 favorable, 4 blocked); no distinct physical mechanism or separate execution is established solely by this row.

15) side.a3: Shared Side-Report Reference (battery-side.a3): Has the same evidential scope as side.a1 and side.a2 (19 favorable, 4 blocked). Treating A1, A2, and A3 as three independent successful experiments would manufacture corroboration from table structure.

16) side.a4: Support and Stability Report (battery-side.a4): Records the alternate-support/stability report (side_a 4.py): 19 stored passes and 4 blocked entries. Report completeness, a favorable numerical finding, and physical support calibration remain distinct, and the eigsh $\mathbf{K}_s^{-1}\mathbf{G}$ symmetry/residual audit remains open.

17) solid.tip: Solid-Result Completeness (battery-solid.tip): Twenty stored passes and three blocked records (tip_solid.py). The 20 stored passes mean that local-solid result records were complete under the column's rules, not 20 favorable strength comparisons.

18) gate.rs_v2: Combined Grouped Finding (battery-gate.rs_v2): Records the combined historical RS-v2 finding (bench.rs): 19 failures and 4 blocked reference entries, with zero passes.

VIII. Complete 44-Entry Catalog Crosswalk (G16, RSV2, and battery)

Every entry in docs/tests/catalog.json (16 G16 legacy predicates, 10 RSV2 requirement groups, and 18 battery columns = 44 total entries) is preserved in the machine-readable crosswalk (JR_EI_P3_T04_atlas_crosswalk_44.json) and summarized below with its exact docs/tests/README.md anchor, original historical Family 1–11 taxonomy (retained below; the current main paper groups the observations into seven rows in Table VII, Section IX rather than numbering eleven families), and visual binding or explicit missing-visual disclosure:

- **Legacy G16 Diagnostics (16 entries):** g16-cad_watertight_and_single_solid (Family 1, export-identity.svg A–B); g16-geometric_and_holder_gates (Family 1, *subsection visual missing: geometry/reference visual not generated*); g16-karambit_ring_and_curved_claw (Family 1, *subsection visual missing: geometry/reference visual not generated*); g16-c6_aesthetic_floor (Family 2, reference.svg A–B); g16-cert_hiker_180n (Family 3, plate-fields.svg B); g16-cert_hunter_260n (Family 3, plate-fields.svg B); g16-strict_hunter_260n (Family 3, assessment.svg C); g16-cert_ring_pull_1500n (Family 4, plate-fields.svg C); g16-twist_body_1p82nm (Family 5, plate-fields.svg D); g16-twist_tip_only_1p82nm (Family 5, plate-fields.svg D); g16-twist_tip_submodel_strict_1p82nm (Family 5, assessment.svg C); g16-solid_tip_3d_p2_1p82nm (Family 6, verification.svg C–D); g16-side_a4_buckling_and_support (Family 7, assessment.svg A); g16-failure_sequence_1p2 (Family 8, verification.svg A); g16-plate_gci_valid_3level (Family 10, verification.svg A–B); g16-dolfinx_cross_check_2pct (Family 10, verification.svg C).
- **Grouped RS-v2 Predicates (10 entries):** rsv2-aesthetic_and_geometric (Family 2, reference.svg A–B); rsv2-floors_828_joint (Family 3/4, plate-fields.svg A–C); rsv2-twist_floor_body_and_tipsub_strict (Family 5, assessment.svg C); rsv2-solid_tip_sf (Family 6, verification.svg C–D); rsv2-strict_hunter_lateral (Family 3, assessment.svg C); rsv2-seq_hard (Family 8, verification.svg A); rsv2-human_max (Family 4/5, assessment.svg B); rsv2-fatigue (Family 9, assessment.svg B); rsv2-gate_s_sharp_law (Family 1, *subsection visual missing: exported-measurement visual not generated*); rsv2-gate_s_tip_point_brep (Family 1, *subsection visual missing: exported-measurement visual not generated*).
- **Historical Battery Evaluators (18 entries):** All 18 battery-* anchors map to battery-matrix.svg (exact evaluator-ID column across all 23 historical subjects), with secondary assessed-example visual links on battery-solid.tip (verification.svg C–D, Family 6), battery-plate.seq (verification.svg A, Family 8), battery-plate.gci (verification.svg A–B, Family 10), battery-side.a4 (assessment.svg A, Family 7), and battery-plate.l7 (assessment.svg B, Family 9).

IX. Missing Subsection Visuals and Unavailable Spatial Fields

To prevent visual availability from being misread as full spatial-field coverage ('40/44' catalog entries have subsection visual links; '6/44' link to the four retained $0.25\,\mathrm{mm}^2$ plate-field arrays in plate-fields.svg), we preserve two explicit non-availability ledgers from visuals.rs:

1) **Four Unpictured Geometry/Export Subsection Entries (subsection_visual_missing):**
   - g16-geometric_and_holder_gates and g16-karambit_ring_and_curved_claw: *"Subsection geometry/reference visual not generated; original flags are retained separately, without an admitted visual comparison."*
   - rsv2-gate_s_sharp_law and rsv2-gate_s_tip_point_brep: *"Subsection exported-measurement visual not generated; nominal geometry and measured export observations must remain distinct."*
2) **Four Unavailable Spatial Field Requests (status:archive_required):**

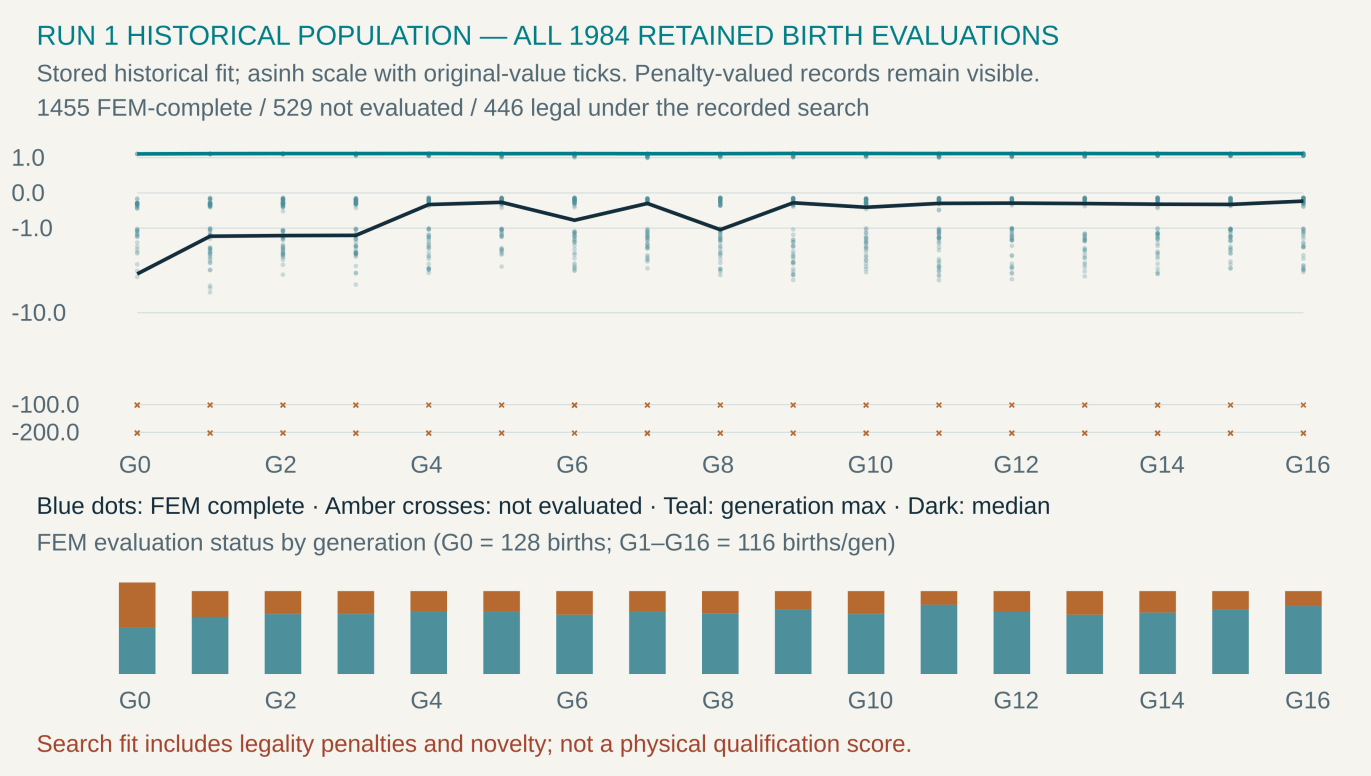


Fig. 17. Supplement Cohort 1 (Run 1) historical birth observations: all 1,984 retained births across generations 0–16 (1,455 FEM-complete blue points, 529 unevaluated amber crosses, 446 search-legal births) on an $\operatorname{asinh}(\text{fit})$ axis. Separate retained membership records additionally contain 2,176 four-island slot memberships, including 192 elite reappearances and 12 ring migrations at generations 4, 8, and 12; these contextual records are not plotted in this figure.

- strict_plate_field: Only certification/strict scalar summaries ($168.9008\,\mathrm{N}$ vs. $168.1786\,\mathrm{N}$ lateral; 0.5239 vs. 0.5287 local torsion) are retained in the admitted publication input set; the $0.25\,\mathrm{mm}^2$ coarse plate field cannot substitute for a $0.0625\,\mathrm{mm}^2$ spatial array.
- solved_solid_field: Only matched scalar safety-factor pairs and 1D longitudinal binned profiles are retained for the P_2 tetrahedral tip submodel; no full 3D volumetric stress array is archived in the publication inputs.
- buckling_mode: Only the historical producer-reported scalar 7.0351 and support schematic are retained. The scalar is not admitted here as a buckling eigenvalue or performance margin; no spatial eigenmode field is retained. Operator/eigenpair validation and dependent qualification inference remain unestablished (Section IV).
- fatigue_field: Only the saved modified-Goodman scalar ($n_f = 0.4718$) is retained; no per-element fatigue heatmap or physical cyclic life measurement exists in the admitted archive.

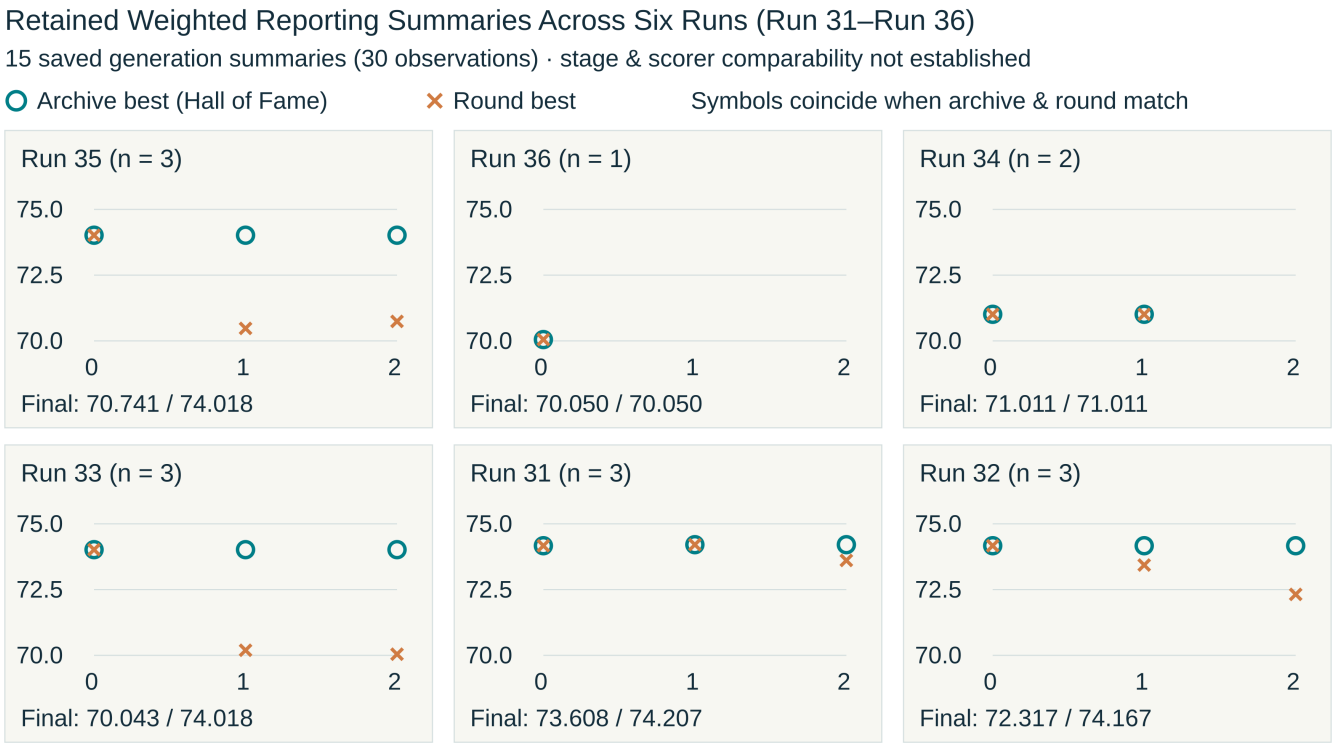


Fig. 18. Supplement Cohort 2 Retained Weighted Reporting Profile (Run 31–Run 36): all fifteen admitted generation summaries across six reporting series (70–75 vertical scale; hollow teal circles show stored archive values, orange crosses show round bests), peaking at $F = 74.207299$ for Individual 2002 in Run 31.

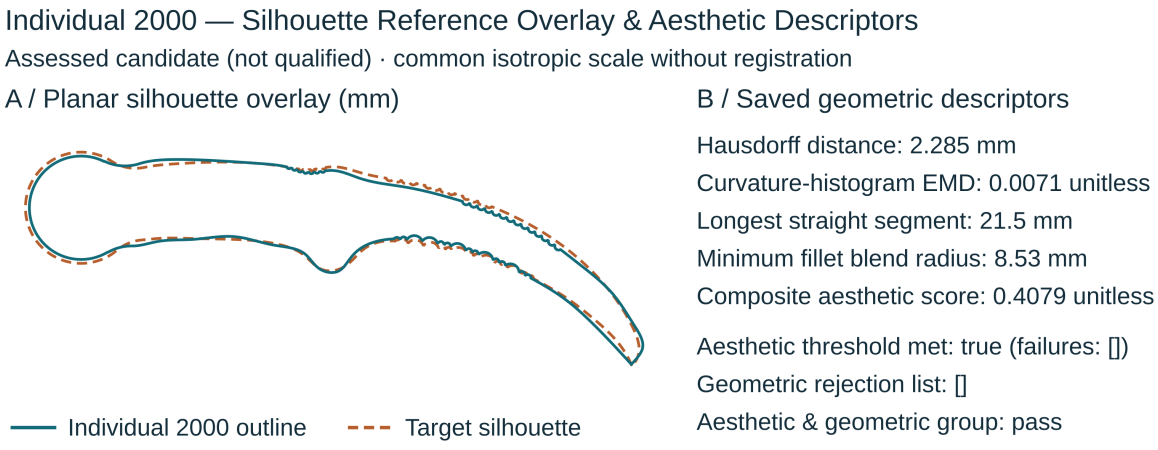


Fig. 19. Supplement Individual 2000 reference-contour comparison records (single assessed export, 1 assessed, 0 qualified): all $4{,}230$ exterior outline points and $1{,}117$ C.5/C.6 reference points displayed as closed polylines under a common isotropic scale, paired with saved descriptors (Hausdorff distance $2.285\,\mathrm{mm}$, curvature EMD 0.0071, similarity score 0.4079).

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